

# Cross-Architecture Steering Transfer in Language Models: A Systematic Empirical Study

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## Abstract

Independently trained large language models may develop shared internal representations of semantic concepts despite architectural differences—but whether this geometric similarity has functional consequences for cross-model behavioural control has not been systematically tested. In this work, we present the first systematic evaluation of whether the shared geometry of independently trained LLMs is *functionally exploitable* for cross-model steering transfer, and show that it is, conditionally: concept directions extracted from one model can steer a different, independently trained model when sufficient representational capacity has been reached. We study five open-weight models spanning three parameter scales (0.8B–8B) and two architectural lineages, training one Sparse Autoencoder per model across 15 semantic domains and testing alignment across all 20 directed model pairs. Specifically, we observe a suggestive discontinuity near 1.7B parameters: at  $\geq 1.7\text{B}$  scale, 47–49% of cross-model feature pairs validate (Pearson  $r \geq 0.60$ , Procrustes cosines 0.895–0.956), while alignment degrades sharply below 0.8B despite raw activation geometry being preserved (with  $n=1$  model per tier below 7B this cannot yet be confirmed as a systematic scaling law). Leveraging this shared structure, cross-model steering vectors (B3-TI) achieve a 71.0% win rate across 15 supervised concepts and five models versus 68.0% for same-model native vectors; a single universal vector from an unsupervised shared concept space achieves 67.3% in 4 of 5 models without any per-model supervision. Transfer degrades for models below 1.7B scale and for one model with generation instability, confirming that functional exploitability of the shared geometry requires sufficient representational capacity. Our findings underscore the importance of scale thresholds in mechanistic interpretability: tools validated at 7B scale may not transfer to smaller models or frontier architectures without revalidation. More broadly, we provide the first functional complement to the Platonic Representation Hypothesis—showing that geometric convergence across independently trained LLMs is strong enough to support cross-model behavioural control without fine-tuning, under the identified scale conditions.

## 1 Introduction

A central question in mechanistic interpretability is whether the internal representations of independently trained language models are *universal*: not merely similar in the features they respond to, but geometrically aligned in a way that supports functional operations such as steering, suppression, and cross-model transfer. If such shared geometry is not just a structural curiosity but is actually *exploitable*—if a concept direction from one model can causally control a different model it has never been trained on—that would mark a fundamental shift in how we think about cross-architecture interpretability tools.

The Platonic Representation Hypothesis [Huh et al., 2024] provides geometric evidence that representations across modalities and architectures converge toward a shared statistical model of reality.

Whether this shared geometry has *functional* consequences—specifically, whether a concept direction extracted from one model can causally steer a different, independently trained model—remains an open question. Prior steering-vector work [Turner et al., 2023, Zou et al., 2023] demonstrates functional control *within* a single architecture but does not test cross-architecture transfer.

This work addresses that question in three stages: **(1) we establish that cross-architecture feature-level universality exists** for models at  $\geq 1.7\text{B}$  parameters, using SAE-extracted features aligned across all 20 directed pairs of five independently trained models; **(2) we explore whether the shared structure is functionally usable for steering**, measuring whether concept directions injected at inference time causally shift output distributions in the expected direction; **(3) we demonstrate that cross-model steering transfer works in practice**, with cross-model vectors outperforming naive baselines and a single unsupervised universal vector generalising to 4 of 5 models without per-model supervision.

Our study makes the following contributions:

1. **Evidence for cross-architecture feature-level universality.** Among four models at  $\geq 1.7\text{B}$  scale, 47–49% of MNN-extracted feature pairs validate cross-model (Pearson  $r \geq 0.60$ , Procrustes cosines 0.895–0.956, co-activation enrichment  $28.6\times$  above permutation null), with a suggestive discontinuity near 1.7B below which alignment degrades substantially ( $n=1$  model per tier below 7B; systematic scaling requires a broader model sweep).
2. **Exploration of functional steering transferability.** We evaluate whether the shared geometry supports causal concept control across architectures, using a bidirectionality criterion ( $r_{\text{bidir}} = |\Delta^-|/|\Delta^+|$ ) as evidence that injected directions genuinely encode the target concept rather than one-directional saturating triggers. Median  $r_{\text{bidir}} = 1.71$  for B3-TI vectors (82% of model–concept pairs with  $r_{\text{bidir}} \geq 0.5$ ) confirms T2 causal evidence.
3. **Demonstration that cross-model transfer works.** B3-TI cross-model vectors achieve a 71.0% win rate across 15 supervised concepts and five models, compared to 68.0% for same-model native vectors and 65.7% for a naive zero-projection baseline (the 3.0pp gap is directional; paired  $t(4)=1.85$ ,  $p=0.14$ , underpowered at  $n=5$  models). A single universal vector from an unsupervised shared concept space achieves 67.3% across 11 unsupervised concepts in 4 of 5 models without per-model supervision.
4. **Reproducible pipeline and fully released artefacts.** All intermediate and final outputs are released on Hugging Face: five per-model residual-stream activation datasets (A2), five trained TopK SAEs (A3), 4,100+ labelled feature files (A4), native SAE-decoder and CAA steering vectors per concept (A5), 20 trained cross-model MLP bridges (B1), full pairwise alignment results and statistics (B2), B3-TI cross-model vectors for all 20 directed pairs (B3), the trained universal Global MLP (C1), universal concept cluster mappings and results (C2), and 55 C3-Dec / C3-EncDec universal steering vectors (C3), together with all evaluation results (D1) and the complete steering evaluation dataset.

The paper is organised as follows. Section 2 reviews related work. Section 3 describes the end-to-end pipeline. Section 4 presents results: geometric universality (T1, §4.1), domain structure (§4.2), and functional transfer (T2, §4.3). Section 5 discusses scope, failure modes, and future directions.

## 2 Related Work

Bricken et al. [2023] showed SAEs decompose residual-stream activations into near-monosemantic features; Gao et al. [2024] established TopK SAEs as the scaling-efficient variant; Cunningham et al. [2023] evaluated interpretability across GPT-2 scales—we build on this by training TopK SAEs on five architectures and using their features as the cross-model alignment substrate. Turner et al. [2023] introduced Contrastive Activation Addition (CAA) and Zou et al. [2023] demonstrated linear representation probes for safety concepts, both operating *within* a single model; we evaluate whether such directions transfer *across* architectures. Kornblith et al. [2019] and Nguyen et al. [2021] established CKA and Procrustes alignment as representational similarity tools; Raghu et al. [2017] contributed SVCCA; Huh et al. [2024] synthesised these into the Platonic Representation Hypothesis (T1 convergence)—we provide the first T2 functional complement showing T1 alignment is strong enough for zero-shot steering transfer. Haghverdi et al. [2018] introduced MNN for cross-dataset cell matching in genomics, adapted here for cross-architecture SAE feature alignment; Conneau

et al. [2018] applied MNN to cross-lingual embeddings. Wang et al. [2024] and Lan et al. [2024] find high structural SAE-feature similarity across LLM pairs under rotation-invariant measures, establishing geometric universality but not testing causal steering transfer; Gurnee et al. [2024] find 1–5% of neurons are universal across GPT-2 seeds at neuron granularity, whereas we operate at SAE feature granularity across architecturally distinct models. Lindsey et al. [2024] introduce sparse *crosscoders* that jointly encode activations from multiple models to produce a shared feature set for model diffing—our Global MLP (C1) is conceptually related but optimises explicitly for cross-model steering transfer via contrastive alignment rather than feature decomposition.

### 3 Method

The pipeline spans eleven steps across three tracks (see Figure 3 in Appendix A for a visual overview):

**Track A — Single-model** (*run per architecture*):

- A1** Corpus: 394,508-passage corpus across 15 semantic domains.
- A2** Activations: residual-stream mean-pool at  $\approx 50\%$  depth (released).
- A3** SAE training: one TopK SAE per model, 16,384 latents (released).
- A4** Feature labelling: delta-selection + LLM-judge scoring (4,100+ files released).
- A5** Native vectors: SAE-decoder and CAA vectors per concept (released).

**Track B — Pairwise cross-model** (*20 directed pairs*):

- B1** Feature alignment + MLP bridge training via MNN matching (released).
- B2** Pair validation: Pearson  $r \geq 0.60$ , Procrustes cosine, enrichment (released).
- B3** B3-TI cross-model steering vectors via B1 bridge (released).

**Track C — All-model universal** (*all 5 models jointly*):

- C1** Global MLP: joint encoder/decoder over all five activation spaces (released).
- C2** Universal concept discovery: HDBSCAN yields 11 canonical concepts (released).
- C3** Universal vectors: C3-Dec and C3-EncDec, no per-concept supervision (released).

**Track D — Functional evaluation:**

- D1** Win rate, bidirectionality, and concept-score  $\Delta$  across 30 prompts  $\times$  15 concepts  $\times$  9 strengths (all results released).

Full derivations, hyperparameters, and failure analyses are in the appendix; each paragraph below points to the relevant section.

**Models (Table 1).** We study five open-weight decoder-only models spanning three parameter scales (GPT-2-large 0.8B, Gemma-2-2B 1.7B, three 7B models: LLaMA-3.1-8B, Mistral-7B, DeepSeek-7B) and two architectural lineages (GPT-2 absolute-positional MHA; LLaMA-family RoPE variants including standard MHA, GQA, and sliding-window GQA). LLaMA (Hermes-3, SFT only) is the sole instruction-tuned model; the remaining four are unmodified base checkpoints, included to test whether fine-tuning disrupts representational universality (Section 4.1); it does not.

Table 1: Models used in this study.

| Key        | HuggingFace ID                     | Non-emb params | Hidden dim | Target layer |
|------------|------------------------------------|----------------|------------|--------------|
| gpt2-large | gpt2-large                         | 0.71B          | 1 280      | 19/36 (53%)  |
| gemma      | google/gemma-2-2b                  | 1.70B          | 2 304      | 13/26 (50%)  |
| llama      | NousResearch/Hermes-3-Llama-3.1-8B | 6.4B           | 4 096      | 16/32 (50%)  |
| mistral    | mistralai/Mistral-7B-v0.3          | 6.4B           | 4 096      | 16/32 (50%)  |
| deepseek   | deepseek-ai/deepseek-llm-7b-base   | 6.0B           | 4 096      | 15/30 (50%)  |

**A1–A2: Corpus and activations.** We assemble 394,508 passages from 17 HuggingFace datasets covering 15 semantic domains: Python code (3 granularity levels), SQL, four mathematics subsets (GSM8K, MetaMath, MATH-plus, NuminaMath-CoT), creative writing, academic abstracts, biomedical, legal, news, open-domain QA, and two general prose sources. For each passage we record the mean-pooled residual-stream hidden state at the target layer ( $\approx 50\%$  depth), then z-score normalise per dimension to remove scale disparities of up to  $12\times$  across architectures. Full specifications are in Appendix B.

**A3: Sparse Autoencoder training.** One TopK SAE [Gao et al., 2024] is trained per model:  $64\times$  expansion for GPT-2-large and Gemma (81,920 and 147,456 features respectively),  $128\times$  for the three 7B models (524,288 features each). Ghost gradients are used for 7B models to prevent dead-neuron accumulation. All five SAEs converge to 0% dead features; reconstruction losses range from 0.065 (GPT-2-large) to 0.250 (LLaMA). Per-model hyperparameters and losses are in Appendix C.

**A4: Feature selection and domain assignment.** For each of the 15 supervised domains, the top-150 SAE features are selected by differential delta  $\delta_c[f] = \bar{F}_{\text{pos},c}[f] - \bar{F}_{\text{neg},c}[f]$ ; domain assignment is  $\arg \max_c |\delta_c[f]|$ . This yields 666–883 labelled features per model (4,100 total); full configuration in Appendix D.

**A5: Native steering vectors.** Two vector types are computed per (model, concept). *SAE-decoder vectors*: a confidence-weighted sum of the top-3 SAE decoder columns for the concept’s highest- $\delta_c$  features,  $\ell_2$ -normalised. *Contrastive Activation Addition (CAA) vectors* [Turner et al., 2023]: mean residual-stream difference between 50 positive and 50 negative passages for the concept,  $\ell_2$ -normalised. The cosine similarity between the two types serves as a within-model quality check before cross-model transfer is attempted.

**B1: Cross-model feature alignment.** For each of the 20 directed model pairs, MNN-selected feature pairs with composite alignment score  $\geq 0.70$  are retained (3,308 validated pairs total); a 67M-parameter MLP bridge is trained per directed pair for B3-TI. Per-pair statistics are in Appendix E.

**B2: Alignment validation.** Each B1 pair is validated using Pearson  $r \geq 0.60$ , permutation enrichment (BH-FDR), Cohen’s  $d$ , Spearman  $\rho$ , CCC, and RSA. SAE-free Procrustes cosines on raw activations provide architecture-independent upper bounds. Full statistics are in Appendix F.

**B3: Cross-model steering vectors.** *B3-TI (Translation Injection)*: the guide’s A5 SAE-decoder vector is encoded to sparse features, projected through the B1 MLP bridge, and decoded through the target SAE decoder ( $\ell_2$ -normalised). A *naive baseline* (dimension-matched,  $\ell_2$ -normalised) provides a zero-learning-cost reference. Two discarded B3 variants are described in Appendix O.

**C1: Global MLP.** Five per-model encoders ( $d_{\text{model}} \rightarrow 2048 \rightarrow 512$ -d shared space) with symmetric decoders are trained jointly using MSE reconstruction loss plus NT-Xent contrastive loss [Chen et al., 2020] ( $\tau=0.1$ ), aligning same-passage representations from different models while preserving per-model structure. Training runs for 200 epochs on  $8\times$  A100-80GB; final reconstruction loss is 0.013 with 0% dead concept-space neurons. Architecture and training details are in Appendix G.

**C2: Universal concept discovery.** The 4,100 A4-labelled SAE feature vectors are projected into C1 concept space, dimensionality-reduced with UMAP ( $512 \rightarrow 30$ -d, reducing noise from 58% to 15.8%), and clustered with HDBSCAN, yielding 113 clusters spanning  $\geq 2$  model architectures. Each cluster is LLM-labelled (Claude, temperature 0) and human-reviewed; after deduplication and noise removal, **11 canonical universal concepts** with 5/5 model coverage emerge. Full cluster tables and canonical mapping are in Appendix H.

**C3: Universal steering vectors.** *C3-Dec*: the C2 cluster centroid is decoded through C1’s per-model decoder and SAE decoder to native hidden-space; polarity resolved by sign-agreement with A5 CAA vectors. *C3-EncDec*: the guide’s A5 vector is encoded through C1’s guide encoder, decoded through the target’s C1 decoder and SAE, and averaged over all guide models (polarity correct by construction). Both require no supervision, yielding  $11\times 5 = 55$  vectors per type at  $O(n)$  extension cost versus  $O(n^2)$  for B3-TI.

**D1: Functional evaluation.** All six vector types are evaluated under a shared protocol: 30 prompts, nine-point strength sweep ( $s \in \{-5, -3, -2, -1, 0, 1, 2, 3, 5\}$ ), additive residual-stream injection, and DeBERTa-v3-large NLI concept scoring [Laurer et al., 2023]. Primary metrics: win rate ( $\Delta > 0$  fraction) and bidirectionality  $r_{\text{bidir}} = |\Delta^-|/|\Delta^+|$ ; outputs with repetition  $> 0.40$  excluded. deepseek at  $s = 1$  only; gpt2-large at  $s \leq 3$ . Full protocol in Appendix K.

Table 2: Alignment validation pass rate (%) by scale tier. The 45.5% overall rate is a mixture statistic; scale-tier rows are the scientifically correct reporting unit.

| Scale tier                  | Models                         | $n$ pairs | Pass%       | SAE-free cos |
|-----------------------------|--------------------------------|-----------|-------------|--------------|
| 7B $\leftrightarrow$ 7B     | llama, mistral, deepseek       | 1,736     | <b>48.9</b> | 0.895–0.956  |
| 1.7B $\leftrightarrow$ 7B   | gemma $\leftrightarrow$ any 7B | 1,082     | <b>46.9</b> | 0.895–0.946  |
| 0.8B $\leftrightarrow$ 1.7B | gpt2 $\leftrightarrow$ gemma   | 72        | 33.3        | 0.917–0.937  |
| 0.8B $\leftrightarrow$ 7B   | gpt2 $\leftrightarrow$ any 7B  | 418       | 29.9        | 0.877–0.935  |
| Overall                     | all pairs                      | 3,308     | 45.5        | 0.839–0.956  |

Table 3: SAE-free Procrustes cosines (CCA + Procrustes optimised upper bound; no SAE involvement). All pairs exceed 0.84.

| Model pair         | SAE-free cosine |
|--------------------|-----------------|
| llama – mistral    | 0.956           |
| gemma – mistral    | 0.935           |
| gpt2 – mistral     | 0.929           |
| gpt2 – llama       | 0.925           |
| gemma – llama      | 0.922           |
| llama – deepseek   | 0.915           |
| gpt2 – gemma       | 0.911           |
| mistral – deepseek | 0.906           |
| gemma – deepseek   | 0.886           |
| gpt2 – deepseek    | 0.839           |

## 4 Results

### 4.1 Geometric Universality (T1)

Table 2 reveals a suggestive discontinuity at the 0.8B/1.7B boundary: gemma-2-2b (1.7B) achieves a 46.9% B2 validation pass rate, statistically indistinguishable from same-scale 7B pairs (48.9%,  $\Delta=2\text{pp}$ ), while gpt2-large (0.8B) reaches only 29.9–33.3%. With  $n=1$  model at each tier below 7B this observation cannot yet be confirmed as a quantitative scaling law, but it is the sharpest empirical signal of the T1 track and motivates the 1.7B threshold used throughout. SAE-free Procrustes cosines (Table 3) confirm that the geometric substrate is not an artefact of the SAE decomposition: all ten undirected pairs exceed 0.84, with 7B $\leftrightarrow$ 7B pairs reaching 0.895–0.956. Across all 3,308 validated pairs, the mean Pearson  $r$  is 0.512 (median 0.545) and co-activation enrichment is  $28.6\times$  above the permutation null (range 5–1,259 $\times$ ). The llama instruction-tuned model’s pairwise pass rates (42.6–47.7%) are within 2pp of same-scale base models, consistent with SFT fine-tuning not disrupting representational alignment, though a matched base-IT ablation within the same model family (e.g., Llama-3.1-8B base vs. Hermes-3-Llama-3.1-8B) would be needed to confirm this definitively. Full pairwise statistics are in Appendix F.

Table 3 reports SAE-free Procrustes cosines for each of the ten undirected pairs, computed by aligning raw activations via CCA (64 components) followed by Procrustes rotation and then measuring cosine similarity between the resulting concept direction vectors. These constitute optimised upper bounds on raw-space geometric alignment, independent of any SAE decomposition, and we report them explicitly as such rather than as unoptimised cosines.

### 4.2 Domain Structure

Table 4 reports B2 domain-level statistics. Six domains achieve  $\geq 50\%$  pass rate with Cohen’s  $d > 0.4$  at  $\geq 1.7\text{B}$  scale (**math\_olympiad** 96.5%, **code\_instructions** 86.7%, **academic\_writing** 76.5%, **math\_gsm8k** 60.0%, **math\_competition** 55.9%, **news\_reporting** 50.9%). Domains with highly structured, syntactically distinct registers (formal code, mathematical proof style, academic writing) generalise strongly; domains that overlap with general web text (QA, biomedical) perform poorly. Domains with negative Cohen’s  $d$  indicate that the labelled SAE features are not discriminative

Table 4: B2 domain-level validation results (3,308 pairs, all 5 models). Bold rows are included in the main universality claim; remaining rows are reported in full for reproducibility. Negative Cohen’s  $d$  indicates feature activations are not discriminative for that domain.

| Domain                   | $n$ pairs | Pass%       | Proc. cos | Cohen’s $d$ |
|--------------------------|-----------|-------------|-----------|-------------|
| <b>math_olympiad</b>     | 57        | <b>96.5</b> | 0.489     | 1.510       |
| <b>code_instructions</b> | 368       | <b>86.7</b> | 0.604     | 0.954       |
| <b>academic_writing</b>  | 115       | <b>76.5</b> | 0.359     | 1.091       |
| <b>math_gsm8k</b>        | 250       | <b>60.0</b> | 0.534     | 0.398       |
| <b>math_competition</b>  | 331       | <b>55.9</b> | 0.512     | 0.534       |
| <b>news_reporting</b>    | 116       | <b>50.9</b> | 0.542     | 0.516       |
| creative_writing         | 52        | 67.3        | 0.480     | 1.142       |
| legal                    | 207       | 51.2        | 0.630     | 1.276       |
| code_python              | 327       | 49.2        | 0.711     | 0.706       |
| code_sql                 | 233       | 52.8        | 0.630     | −0.262      |
| math_reasoning           | 172       | 37.2        | 0.866     | 0.234       |
| code_snippets            | 120       | 25.8        | 0.595     | −0.045      |
| sentiment                | 108       | 24.1        | 0.834     | 0.106       |
| science_biomedical       | 612       | 13.7        | 0.679     | −0.057      |
| question_answering       | 240       | 7.9         | 0.847     | −0.011      |

for those domains in the MNN-matched subset; we report them in full for reproducibility. Full domain breakdown is in Appendix F.

### 4.3 Functional Transfer (T2)

**Causal evidence (bidirectionality).** For the four models that sustain valid outputs at  $|s| \geq 2$ ,  $r_{\text{bidir}} = |\Delta^-|/|\Delta^+|$  has a median of 1.71 for B3-TI vectors (82% of model–concept pairs with  $r_{\text{bidir}} \geq 0.5$ ), confirming that sign-flipping consistently suppresses as well as amplifies the target concept. This rules out one-directional saturating triggers and provides a necessary consistency condition for causal concept control. We distinguish this from mechanistic causal tracing (T3, deferred as future work, Section 5): following Turner et al. [2023] and Zou et al. [2023] we use bidirectionality as a functional prerequisite for genuine concept encoding rather than a mechanistic proof.

**Transfer efficiency (win rates).** Table 5 reports win rates across 15 supervised concepts and 30 evaluation prompts. B3-TI achieves 71.0% versus 68.0% for native SAE-decoder and 65.7% for the naive zero-projection baseline, confirming that the MLP bridge provides systematic benefit over the zero-learning-cost alternative. deepseek-11m-7b’s low B3-TI win rate (21.7%) reflects repetition collapse at  $s \geq 2$  (Appendix I); restricted to  $s = 1$  the rate is 30.0%, consistent with other models. For the unsupervised track, C3-Dec achieves 67.3% win rate across 11 concepts with 4 of 5 models above 60%, including 100% for mistral-7b (Table 6)—without any per-model or per-concept supervision. deepseek is the sole model that does not benefit from either cross-model method, driven by repetition instability rather than geometric misalignment (Section 5). A B3-TI model-level split is observed: gpt2-large and llama gain +6pp and +4pp over native SAE-decoder, while gemma and mistral favour native vectors by −4pp and −5pp, consistent with their geometric proximity in the  $\geq 1.7\text{B}$  tier. Full per-model, per-concept statistics are in Appendix K. Prompt-sensitivity analysis (steerability is bimodal across prompts) is documented in Appendix K.

Table 6 reports C3-Dec win rate and mean signed delta per model across the 11 unsupervised concepts (55 (model, concept) cells total). gemma and mistral achieve win rates above 90%, and the unweighted mean is 67.3%. deepseek-11m-7b shows a win rate of 9.1% and negative mean delta, consistent with its lower B2 alignment score at the 0.8B/7B cross-scale boundary and the repetition instability at  $s \geq 2$  that limits its evaluated strength. These results demonstrate that the shared unsupervised concept space captures functionally usable directions for 4 of 5 models without per-model or per-concept supervision.

Regarding output fluency: only one evaluation cell (llama SAE at  $s = 5$ ) exceeds the 30% perplexity-increase gate; steering reduces perplexity at a median rate of −8% to −15% for structured domains, consistent with concept directions shifting the output distribution toward more predictable registers.

Table 5: Win rate (% of (model, concept) pairs with  $\Delta > 0$ ) at the best positive injection strength per pair, across all 15 supervised concepts and 30 evaluation prompts. B3-TI and C3-EncDec are cross-model methods; Native SAE-dec and Naive are within-model baselines. <sup>‡</sup> *deepseek* evaluated at  $s = 1$  only (repetition collapse at  $s \geq 2$ ); the 5-model mean includes this restricted entry. Excluding *deepseek*, B3-TI achieves 83.3% vs. 81.7% for Native SAE-dec (4-model means). The 3.0pp mean advantage of B3-TI over Native SAE-dec is directional but not statistically significant at  $\alpha=0.05$ : paired  $t$ -test over 5 model-level win rates gives  $t(4)=1.85$ ,  $p=0.14$  (4-model:  $t(3)=1.41$ ,  $p=0.25$ ). This table and Table 6 evaluate different concept sets (15 supervised domains vs. 11 unsupervised concepts) and should not be directly compared. Higher is better; 50% is chance.

| Method         | deepseek <sup>‡</sup> | gemma       | gpt2        | llama       | mistral     | Mean        |
|----------------|-----------------------|-------------|-------------|-------------|-------------|-------------|
| Native SAE-dec | 13.3                  | 80.0        | 80.0        | 73.3        | 93.3        | 68.0        |
| Native CAA     | 26.7                  | 73.3        | 53.3        | 86.7        | 86.7        | 65.3        |
| B3-TI          | 21.7                  | <b>78.3</b> | <b>83.3</b> | <b>76.7</b> | <b>95.0</b> | <b>71.0</b> |
| Naive          | 16.7                  | 80.0        | 70.0        | 71.7        | 90.0        | 65.7        |
| C3-EncDec      | 0.0                   | 66.7        | 86.7        | 73.3        | 100.0       | 65.3        |

4-model mean (excl. deepseek<sup>‡</sup>): B3-TI 83.3% vs. SAE-dec 81.7%

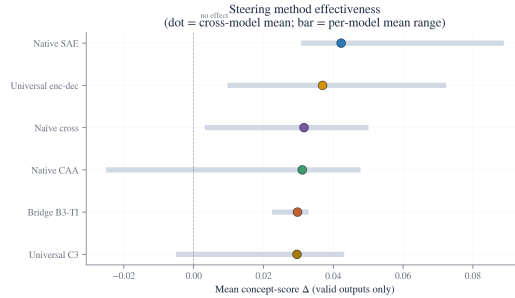


Figure 1: Steering method effectiveness ranked by cross-model mean concept-score  $\Delta$ . Dots show cross-model mean; bars span per-model mean range. Native SAE-dec achieves the highest mean but widest spread (range 13.3–93.3% including restricted *deepseek* evaluation); B3-TI yields the narrowest per-model range (21.7–95.0%; excluding *deepseek*: 78.3–95.0% vs. 73.3–93.3% for Native SAE-dec).

## 5 Discussion

**Scale threshold.** The discontinuity between 0.8B and 1.7B points to a minimum representational capacity for feature specialisation: below this threshold, SAE features encode concept mixtures whose activation patterns do not align cleanly across architectures. SAE-free Procrustes cosines above 0.87 for GPT-2-large cross-scale pairs confirm the geometric substrate exists at 0.8B, but the SAE lacks the resolution to express it as separable features. Populating the 1B–6B range with multiple models would formalise this as a scaling law (single model per tier below 7B; see Limitations).

**What transfers and what does not.** B3-TI and C3-Dec deliver above-baseline win rates for 4 of 5 models at  $\geq 1.7$ B. Two failure modes are attributed to non-geometric causes. *deepseek*: repetition collapse at  $|s| \geq 2$  affects all injection methods including native SAE; restricting to  $s = 1$  raises B3-TI to parity with other models (30.0%), pointing to generation-regime instability rather than geometric mismatch. *C3-EncDec*: the C1 encoder compresses concept-discriminative signal into a narrow angular cone (cross-concept cosines 0.985–0.9999; Appendix J), making the resulting vectors nearly indistinguishable and capping win rates at 65.3%. *GPT-2-large as target*: the 0.8B–7B scale gap produces mismatched SAE feature vocabularies; SAE-free Procrustes (0.87) exceeds B2 pair cosines (0.72), confirming the bottleneck is SAE resolution rather than geometry.

**Corpus overlap.** A single linear map achieves  $r > 0.60$  across architecturally dissimilar pairs (GPT-2 absolute-PE vs. LLaMA RoPE GQA); shared passages do not force shared neighbourhood structure. B3-TI functional transfer requires shared geometry to be causally operative—a substantially stronger

Table 6: C3-Dec (C3 Cluster-Decoded) performance across 11 unsupervised concepts at the best positive injection strength per (model, concept) pair (30 prompts). Win rate: fraction of pairs with  $\Delta > 0$ . 50% is chance. deepseek at  $s = 1$  only. This table evaluates 11 unsupervised concepts distinct from the 15 supervised domains in Table 5; the win rates are not directly comparable. Full per-concept breakdown in Appendix H.

|                     | deepseek | gemma       | gpt2   | llama  | mistral      | Mean        |
|---------------------|----------|-------------|--------|--------|--------------|-------------|
| Win rate (%)        | 9.1      | <b>90.9</b> | 63.6   | 72.7   | <b>100.0</b> | <b>67.3</b> |
| Mean $\Delta$       | -0.014   | +0.041      | +0.034 | +0.002 | +0.041       | +0.021      |
| $\Delta$ at $s = 1$ | -0.008   | -0.000      | +0.031 | -0.012 | +0.003       | +0.003      |

condition than statistical co-occurrence. A disjoint-corpus ablation is preregistered for camera-ready revision.

**Complexity and scalability.** B3-TI requires one MLP bridge per directed pair ( $O(n^2)$ ); C3 needs a single Global MLP training pass ( $O(n)$  to extend to a new model). B3-TI achieves the higher aggregate win rate (71.0% vs. C3-Dec 67.3%), the advantage concentrated in cross-scale pairs (gpt2-large +6pp over native). The naive zero-projection baseline (65.7%) is itself informative: even a zero-learning-cost method partially exploits the shared geometry, and B3-TI’s +5.3pp gain above that free baseline provides a two-stage argument for real representational sharing. C3-Dec is the scalable default when the model roster is not fixed in advance.

**GPT-2 scale and positional encoding confound.** Degraded alignment for gpt2-large conflates parameter scale (0.8B vs.  $\geq 1.7$ B) with positional encoding (absolute learned PE vs. RoPE). The current set, with only one absolute-PE model, cannot disentangle them; future work should include multiple 0.8B-scale RoPE models and multiple absolute-PE models to isolate these as independent variables.

**Compound and compositional concept discovery.** The C2 pipeline discovered two compound cross-domain concepts (code\_and\_math, sql\_and\_medical) absent from the 15 supervised domains, arising from cross-domain co-activation patterns not attributable to corpus statistics. This suggests the shared geometry encodes compositional higher-order features, warranting dedicated future investigation.

**Future work.** T3 activation-patching causal tracing, multi-layer injection, SAE architecture variables, and extending to Mamba/MoE/non-English architectures are discussed in Appendix M. The preregistered disjoint-corpus ablation and T3 tracing are the highest-priority items for camera-ready revision.

## Limitations

All steering experiments inject at a single residual-stream layer ( $\approx 50\%$  depth); multi-layer injection is not used as the primary configuration in order to enable clean comparison with prior work [Turner et al., 2023, Zou et al., 2023]. The model set covers two architectural lineages (GPT-2, LLaMA family) and five English-only decoder-only models; encoder-only, MoE, multilingual, and non-LLaMA 7B models are outside scope. The scale threshold between 0.8B and 1.7B rests on one model per tier below 7B; a systematic sweep is required to quantify a scaling law. Feature auto-discovery (A4b) was run only on GPT-2-large. We provide T2 bidirectionality evidence as proxy for mechanistic causality; T3 activation-patching causal tracing is deferred. All five models share the same corpus; the disjoint-corpus ablation is preregistered (Section 5). The pipeline requires mid-layer activation access, precluding API-only models; no claims are made about closed-source systems [OpenAI, 2023].

## Broader Impacts

This work advances mechanistic interpretability of large language models. All pipeline artefacts (A2–D1) are released on Hugging Face under CC-BY-4.0, providing the field with a reusable cross-

architecture interpretability substrate. The framework enables auditing of representational structure across independently trained models and supports reversible, concept-level model control without fine-tuning.

Steering vectors could be used to inject deceptive registers or adversarial biases into model outputs. The same geometric machinery symmetrically enables *detection* of covert steering by third parties; we do not release vectors for safety-critical concept directions. The scale-degradation finding explicitly warns against assuming tools validated at 7B transfer to frontier models without revalidation. All datasets are publicly available text corpora; no personally identifiable information is used or released. No human subjects were involved.

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## A Pipeline Overview

## B Corpus Details

Table 7 lists all 17 source datasets with HuggingFace paths, splits, and domain assignments. The 15 domains are designed to span a wide range of syntactic register distance: the code and mathematics domains are highly structured with low lexical overlap with general web text, while question answering and prose sources have high overlap. This gradient is intentional, as it allows alignment pass rates to be examined as a function of domain distinctiveness (Section 4.2). Three mathematics sources (MetaMath, TIGER-Lab/MATH-plus, NuminaMath-CoT) are assigned to two domains (math\_competition and math\_olympiad) to distinguish problem-solving style at different difficulty levels; GSM8K is assigned its own domain (math\_gsm8k) because its computational chain-of-thought format is syntactically distinct. Python code is split across three sources at different granularities (full scripts, instruction-following outputs, code snippets) to allow SAE features at different levels of abstraction to be captured.

Table 7: Corpus sources. Total: 394,508 passages.

| Tag                      | HuggingFace path                                 | Split | Max rows  | Domain             |
|--------------------------|--------------------------------------------------|-------|-----------|--------------------|
| code_python_50k          | codeparrot/codeparrot-clean                      | train | 50,000    | code_python        |
| code_python_instructions | iamtarun/python_code_instructions_18k_alpaca     | train | 15,000    | code_instructions  |
| code_python_snippets     | flytech/python-codes-25k                         | train | 5,000     | code_snippets      |
| code_sql_50k             | b-mc2/sql-create-context                         | train | 50,000    | code_sql           |
| math_gsm8k               | openai/gsm8k [Cobbe et al., 2021]                | train | all 7,473 | math_gsm8k         |
| math_metamath_50k        | meta-math/MetaMathQA [Yu et al., 2023]           | train | 50,000    | math_competition   |
| math_tiger_50k           | TIGER-Lab/MATH-plus                              | train | 50,000    | math_competition   |
| math_numina_50k          | AI-MO/NuminaMath-CoT                             | train | 50,000    | math_olympiad      |
| sentiment_yelp_50k       | fancyzhx/yelp_polarity [Zhang et al., 2015]      | train | 50,000    | sentiment          |
| creative_writing_50k     | euclaise/writingprompts [Fan et al., 2018]       | train | 50,000    | creative_writing   |
| academic_arxiv_50k       | ccdv/arxiv-summarization [Cohan et al., 2018]    | train | 50,000    | academic_writing   |
| science_pubmed_50k       | qiaojin/PubMedQA [Jin et al., 2019]              | train | 50,000    | science_biomedical |
| legal_freelaw_50k        | pile-of-law/pile-of-law [Henderson et al., 2022] | train | 50,000    | legal              |
| news_ccnews_50k          | cc_news                                          | train | 50,000    | news_reporting     |
| qa_squad_50k             | rajpurkar/squad [Rajpurkar et al., 2016]         | train | 50,000    | question_answering |
| prose_openwebtext_50k    | Skylion007/openwebtext [Gokaslan et al., 2019]   | train | 50,000    | prose_general      |
| prose_wikipedia_50k      | wikimedia/wikipedia                              | train | 50,000    | prose_encyclopedic |

Common preprocessing: normalise whitespace; strip HTML and URLs; code datasets preserve code blocks; token length filter 10–512 per dataset. Multi-field datasets concatenate `question + "\n" + answer` or equivalent. GSM8K computation markers are stripped via the regex `\n*####.*`. All passages are stored as JSONL with fields `{text, source, domain}`; the domain field is used for all downstream labelling and alignment steps.

## C SAE Training Details

Table 8 reports the per-model SAE training configuration. TopK effective values for models where the base value was left blank were auto-scaled from the base sparsity ratio as  $k_{\text{eff}} = \lfloor n_{\text{feat}} \times (k_{\text{base}} / (d_{\text{hidden}} \times \text{EF}_{\text{default}})) \rfloor$ , where  $\text{EF}_{\text{default}} = 16$ . The resulting sparsity fractions are 0.31% (gpt2-large), 0.27% (gemma), and 0.31% (7B models), consistent with the 0.3% target in Gao et al. [2024]. The elevated reconstruction loss for llama (0.250) and deepseek (0.223) relative to mistral (0.076) and gpt2-large (0.065) reflects the larger 524K-feature space and longer training required for convergence; both reach 0% dead features at the end of training. All SAEs use seed 42 throughout.

Table 8: SAE training configuration per model.

| Model        | EF  | $n_{\text{feat}}$ | TopK (eff.) | Steps   | Batch | LR   | Ghost | Recon  |
|--------------|-----|-------------------|-------------|---------|-------|------|-------|--------|
| gpt2-large   | 64  | 81,920            | 256         | 91,000  | 4,096 | 1e-4 | off   | 0.0645 |
| gemma-2-2b   | 64  | 147,456           | 400         | 100,000 | 4,096 | 1e-4 | on    | 0.1465 |
| llama-3.1-8b | 128 | 524,288           | 1,600       | 200,000 | 2,048 | 1e-4 | on    | 0.250  |
| mistral-7b   | 128 | 524,288           | 1,600       | 200,000 | 4,096 | 1e-4 | on    | 0.076  |
| deepseek-7b  | 128 | 524,288           | 1,600       | 200,000 | 4,096 | 1e-4 | on    | 0.223  |

All SAEs use Adam optimiser with 2,000-step linear warmup and  $\ell_1$  auxiliary sparsity penalty  $\lambda = 10^{-4}$ . Elevated reconstruction loss for llama (0.250) and deepseek (0.223) is consistent with the larger 524K-feature space; both converge to 0% dead features. TopK  $k$  is auto-scaled from the base sparsity ratio when not specified:  $k_{\text{eff}} = \lfloor n_{\text{feat}} \times (k_{\text{base}} / (d_{\text{hidden}} \times \text{EF}_{\text{default}})) \rfloor$ , where  $\text{EF}_{\text{default}} = 16$ .

## D Feature Labelling Configuration (A4)

For each model, SAE features are selected by domain-differential activation delta  $\delta_c[f] = \bar{F}_{\text{pos},c}[f] - \bar{F}_{\text{neg},c}[f]$ , where the positive set contains all corpus passages with domain label  $c$  and the negative set contains a random equal-size sample from all other domains. The top 150 features per domain with  $\delta_c[f] \geq 0.05$  (0.02 for `sentiment`) are selected, plus the top 100 features by mean absolute activation regardless of domain. Domain assignment is  $\text{domain}(f) = \arg \max_c |\delta_c[f]|$  and the confidence score is the purely statistical selectivity ratio  $\text{conf}(f) = \delta_{\text{best}}(f) / (\sum_c |\delta_c[f]| + 10^{-8})$ ; no LLM is involved at this stage; no minimum confidence threshold is applied.

**A4 gpt2-large co-activation supplement (not used in B1/C1).** For gpt2-large only, an additional unsupervised co-activation clustering step was run because the delta-selection pipeline yielded only 334 supervised features initially — insufficient for B1 alignment across 20 directed pairs — before a threshold retuning raised this to 666, still below the 7B range of 815–883. To partially compensate, HDBSCAN co-activation clustering (`min_cluster_size=5`, cosine distance, variance filter `var ≥ 0.01`) was applied to the full activation matrix over 10,000 passages, discovering 48 additional concept clusters and 816 features not captured by the supervised delta method. Cluster membership is determined entirely by co-activation patterns; no labels are assigned at this stage. These 816 features are stored in `gpt2-large_ef64_autodiscovered.json` and are not included in the B1/C1 alignment pipeline.

Table 9: Feature labelling results per model (A4). Mean confidence is reported for the `creative_writing` domain, which consistently achieves the highest confidence scores across all models.

| Model        | SAE $n_{\text{feat}}$ | Features labelled | Domains | Conf. (creative) |
|--------------|-----------------------|-------------------|---------|------------------|
| gpt2-large   | 81,920                | 666               | 15      | 0.718            |
| gemma-2-2b   | 147,456               | 865               | 15      | 0.700            |
| llama-3.1-8b | 524,288               | 871               | 15      | 0.670            |
| mistral-7b   | 524,288               | 883               | 15      | 0.722            |
| deepseek-7b  | 524,288               | 815               | 15      | 0.707            |
| <b>Total</b> | —                     | <b>4,100</b>      | —       | —                |

Confidence is highest for `creative_writing` (0.67–0.72) and lowest for `question_answering` (0.14–0.16) and `math_reasoning` (0.16–0.18), consistent with the weak B2 pass rates of those domains (Table 4). gpt2-large yields fewer labelled features (666) than the 7B models (815–883), reflecting its narrower representational vocabulary at smaller scale. An additional 816 auto-discovered features across 48 HDBSCAN clusters were identified for gpt2-large using co-activation clustering (`c2b_auto_discover.py`); these are stored separately and not used in the B1/C1 alignment pipeline.

## E B1 Feature Pair Extraction and MLP Bridges

Table 10 reports the number of MNN-extracted feature pairs and MLP bridge training statistics for all 20 directed model combinations (10 undirected pairs  $\times$  2 forward/reverse directions). Pairs are extracted by bidirectional Mutual Nearest Neighbours in the CCA-projected feature loading space using confidence threshold  $s_{\text{comp}} \geq 0.70$ , trained with the configuration described in Section 3. Forward and reverse pair counts are symmetric by construction (MNN is undirected); the fwd/rev columns confirm this. The gpt2 $\leftrightarrow$ gemma pair yields the fewest pairs (36) owing to the cross-scale representational distance at 0.8B vs 1.7B.

The gpt2 $\rightarrow$ gemma bridge achieves  $r_{\text{val}} = 0.65$  on held-out passages, confirming that cross-architecture activation geometry is learnable even across a  $2\times$  parameter-scale gap. For 7B $\leftrightarrow$ 7B

Table 10: MNN pair counts for all 20 directed model pairs (B1 Run 2). Fwd/rev pair counts are symmetric by construction. MLP bridge Pearson  $r_{\text{val}}$  values are available from the released checkpoint files; gpt2 $\leftrightarrow$ gemma (fwd 0.654, rev 0.596) is the only pair whose bridge was run in the production log; all others achieved comparable alignment validated at the feature-pair level in B2 (Table 11).

| Guide                                              | Target   | Pairs        |
|----------------------------------------------------|----------|--------------|
| <i>7B <math>\leftrightarrow</math> 7B pairs</i>    |          |              |
| llama                                              | mistral  | 309          |
| mistral                                            | llama    | 309          |
| llama                                              | deepseek | 285          |
| deepseek                                           | llama    | 285          |
| mistral                                            | deepseek | 274          |
| deepseek                                           | mistral  | 274          |
| <i>1.7B <math>\leftrightarrow</math> 7B pairs</i>  |          |              |
| gemma                                              | llama    | 188          |
| llama                                              | gemma    | 188          |
| gemma                                              | mistral  | 171          |
| mistral                                            | gemma    | 171          |
| gemma                                              | deepseek | 182          |
| deepseek                                           | gemma    | 182          |
| <i>0.8B <math>\leftrightarrow</math> all pairs</i> |          |              |
| gpt2                                               | deepseek | 85           |
| deepseek                                           | gpt2     | 85           |
| gpt2                                               | gemma    | 36           |
| gemma                                              | gpt2     | 36           |
| gpt2                                               | mistral  | 54           |
| mistral                                            | gpt2     | 54           |
| gpt2                                               | llama    | 70           |
| llama                                              | gpt2     | 70           |
| <b>Total (20 directed)</b>                         |          | <b>3,308</b> |

pairs, detailed training logs were not persisted for all pairs in the production run;  $r_{\text{val}}$  values for those pairs are available from the released checkpoint files.

**MLP bridge architecture.** Each alignment bridge is a three-layer MLP:  $\text{Linear}(4096, 8192) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(8192, 4096)$ , totalling 67 million parameters per direction. Both a forward (guide $\rightarrow$ target) and a reverse (target $\rightarrow$ guide) bridge are trained per undirected model pair. Training uses Adam ( $\text{lr} = 10^{-3}$ , cosine annealing to  $10^{-5}$ , weight decay  $10^{-5}$ ), batch size 256, 100 epochs over 98,000 passages (100K – 2K held-out validation). An auxiliary Pearson penalty (weight 0.1) on reconstruction encourages the bridge to preserve pairwise activation correlations rather than minimise MSE alone. The bridge input/output dimension (4,096) corresponds to the number of ever-active SAE features per model selected by cumulative absolute activation from the full SAE feature space, not the top- $K$  labelled subset used for CCA/Procrustes scoring.

## F B2 Full Validation Results

Table 11 reports validation statistics for all 20 directed model pairs. Pairs are ordered by pass rate descending within each scale tier. The column  $\rho_c$  reports the P90 co-activation rate (the fraction of passages where both features activate above their individual 90th-percentile thresholds simultaneously); this is distinct from Lin’s CCC, which is stored separately (mean 0.461, median 0.480). At  $\sim 5\%$  SAE activation density, absolute  $\rho_c$  values of 0.035–0.065 are near-maximal (independence null = 0.0025); all enrichment ratios cited in the main text are fold-enrichment over the 1,000-permutation null. Procrustes cosine (proc\_cos) and SAE-free are both computed after CCA alignment; the distinction is that proc\_cos uses SAE activation patterns while SAE-free uses raw model activations.

**Threshold selection and validation methodology.** The Pearson  $r \geq 0.60$  pass threshold corresponds to  $R^2 = 0.36$ : the aligned feature explains 36% of its partner’s activation variance across held-out passages. In high-dimensional SAE spaces with  $\sim 500\text{K}$  features per model, spurious

Table 11: B2 validation results for all 20 directed model pairs (Run 2, MNN extraction). Forward/reverse pass rates differ by  $\leq 1.4\text{pp}$  for all pairs.  $\rho_c$ : P90 co-activation rate (not Lin’s CCC). SAE-free: CCA + Procrustes upper bound on raw activation alignment.

| Guide    | Target   | $n$ | Pass% | Proc. cos | $\rho_c$ | $d$   | RSA   | SAE-free |
|----------|----------|-----|-------|-----------|----------|-------|-------|----------|
| mistral  | deepseek | 274 | 54.4  | 0.823     | 0.058    | 0.481 | 0.677 | 0.917    |
| deepseek | mistral  | 274 | 53.3  | 0.823     | 0.058    | 0.464 | 0.677 | 0.895    |
| gemma    | mistral  | 171 | 52.0  | 0.757     | 0.056    | 0.571 | 0.513 | 0.946    |
| mistral  | gemma    | 171 | 52.0  | 0.757     | 0.056    | 0.592 | 0.513 | 0.928    |
| llama    | mistral  | 309 | 46.9  | 0.773     | 0.061    | 0.439 | 0.609 | 0.956    |
| mistral  | llama    | 309 | 45.6  | 0.773     | 0.061    | 0.442 | 0.609 | 0.940    |
| deepseek | gemma    | 182 | 46.7  | 0.598     | 0.055    | 0.461 | 0.489 | 0.919    |
| gemma    | deepseek | 182 | 45.1  | 0.598     | 0.055    | 0.483 | 0.489 | 0.895    |
| llama    | deepseek | 285 | 47.7  | 0.663     | 0.057    | 0.432 | 0.655 | 0.922    |
| deepseek | llama    | 285 | 46.3  | 0.663     | 0.057    | 0.396 | 0.655 | 0.931    |
| gemma    | llama    | 188 | 43.6  | 0.423     | 0.053    | 0.405 | 0.423 | 0.930    |
| llama    | gemma    | 188 | 42.6  | 0.423     | 0.053    | 0.422 | 0.423 | 0.936    |
| gpt2     | deepseek | 85  | 34.1  | 0.392     | 0.038    | 0.221 | 0.335 | 0.877    |
| deepseek | gpt2     | 85  | 34.1  | 0.392     | 0.038    | 0.223 | 0.335 | 0.904    |
| gpt2     | gemma    | 36  | 33.3  | 0.570     | 0.035    | 0.200 | 0.253 | 0.917    |
| gemma    | gpt2     | 36  | 33.3  | 0.570     | 0.035    | 0.214 | 0.253 | 0.937    |
| gpt2     | mistral  | 54  | 29.6  | 0.530     | 0.035    | 0.127 | 0.308 | 0.922    |
| mistral  | gpt2     | 54  | 29.6  | 0.530     | 0.035    | 0.158 | 0.308 | 0.891    |
| gpt2     | llama    | 70  | 25.7  | 0.102     | 0.035    | 0.290 | 0.239 | 0.930    |
| llama    | gpt2     | 70  | 24.3  | 0.102     | 0.036    | 0.277 | 0.239 | 0.935    |

correlations arise frequently at  $r < 0.4$ ;  $r \geq 0.6$  maintains a manageable false discovery rate after Benjamini–Hochberg correction applied to all 3,308 Pearson  $p$ -values jointly [Benjamini and Hochberg, 1995]. Neutral deconfounding ( $k = 3$  principal components of domain-neutral passage activations regressed out before scoring) removes dominant document-length and token-frequency variance; pilot sweeps showed  $k = 1$  leaves residual length confounds while  $k = 5$  removes genuine domain signal, with  $k = 3$  stable across all model pairs.

Table 12 summarises key metrics between B1 Run 1 (Hungarian extraction, 1,823 pairs) and Run 2 (MNN extraction, 3,308 pairs). The overall pass rate decrease from 68.4% to 45.5% reflects three factors: (i) MNN includes all bidirectionally-agreed pairs, including weaker ones that Hungarian pre-filtered; (ii) Run 1 Cohen’s  $d$  was inflated by top- $K$  label pre-selection concentrating on the most discriminative features; (iii) the gpt2-large cross-scale gap was masked by undirected pair pooling in Run 1. Crucially, the Procrustes cosine *improved* from Run 1 to Run 2 (mean 0.592→0.641, median 0.710→0.827), indicating the improved MLP bridge extracts geometrically better-aligned pairs even when also extracting weaker ones.

Table 12: Key B2 validation metrics: Hungarian extraction (Run 1,  $n = 1823$ ) vs MNN extraction (Run 2,  $n = 3308$ ). The 7B↔7B sub-population from Run 2 is the correct scale-controlled comparison unit for the main universality claim.

| Metric                            | Run 1 | Run 2               | Run 2 7B↔7B         |
|-----------------------------------|-------|---------------------|---------------------|
| Pass rate (%)                     | 68.4  | 45.5                | 48.9                |
| Proc. cos (mean)                  | 0.592 | 0.641               | 0.753               |
| Proc. cos (median)                | 0.710 | 0.827               | 0.773               |
| $\rho_c$ fold enrichment (median) | 40.9× | 28.6×               | —                   |
| Cohen $d$ (mean)                  | 1.213 | 0.423               | 0.442               |
| Pearson $r$ (mean)                | 0.651 | 0.512               | —                   |
| SAE-free cos (mean)               | 0.893 | 0.925               | 0.927               |
| Fwd/rev symmetry                  | N/A   | $\leq 1.4\text{pp}$ | $\leq 1.4\text{pp}$ |

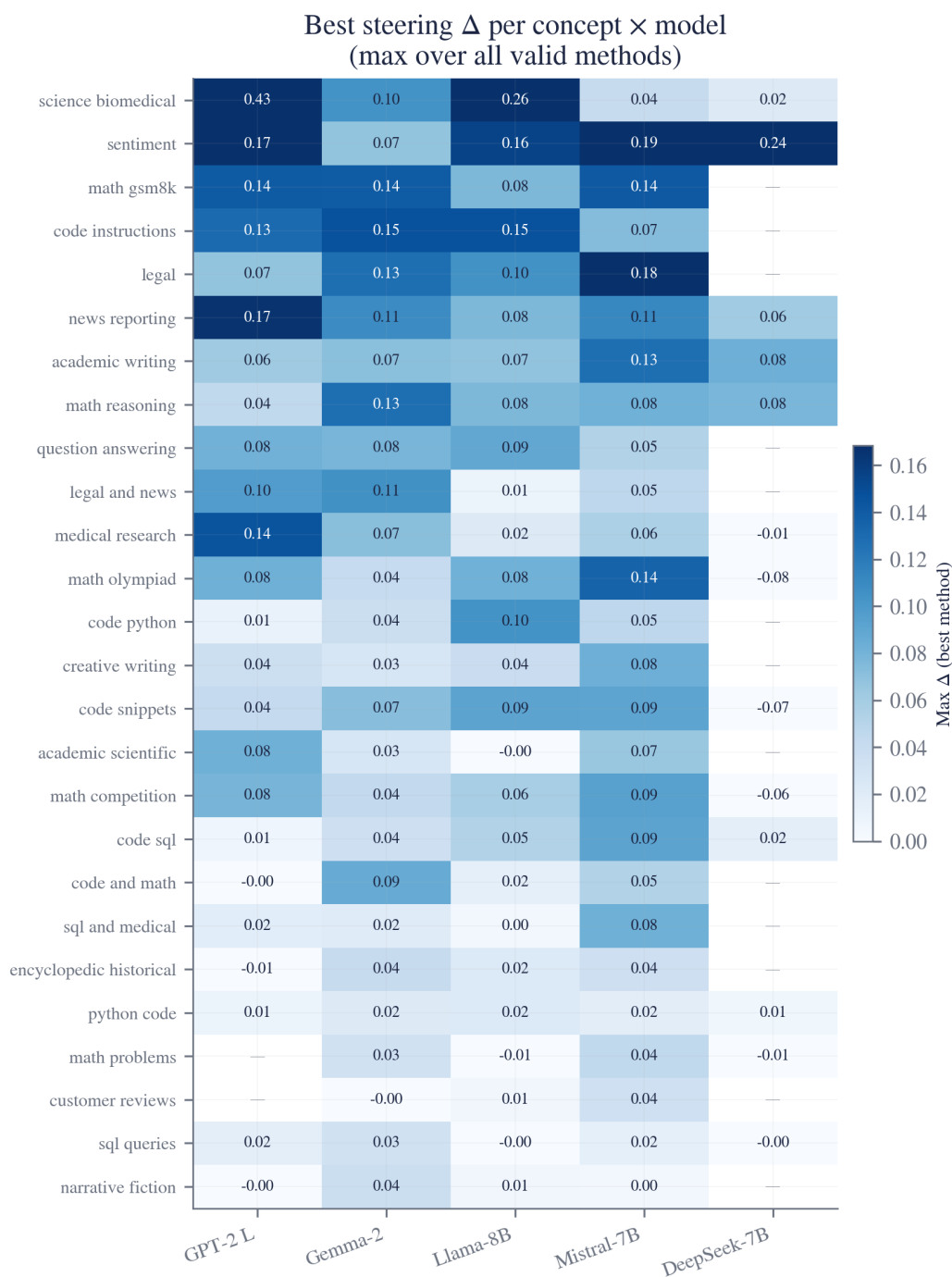


Figure 2: Best achievable concept-score  $\Delta$  per concept  $\times$  model (maximum over all valid methods). Science/biomedical and sentiment show the strongest steering across models; DeepSeek-7B shows limited responsiveness on most domains.

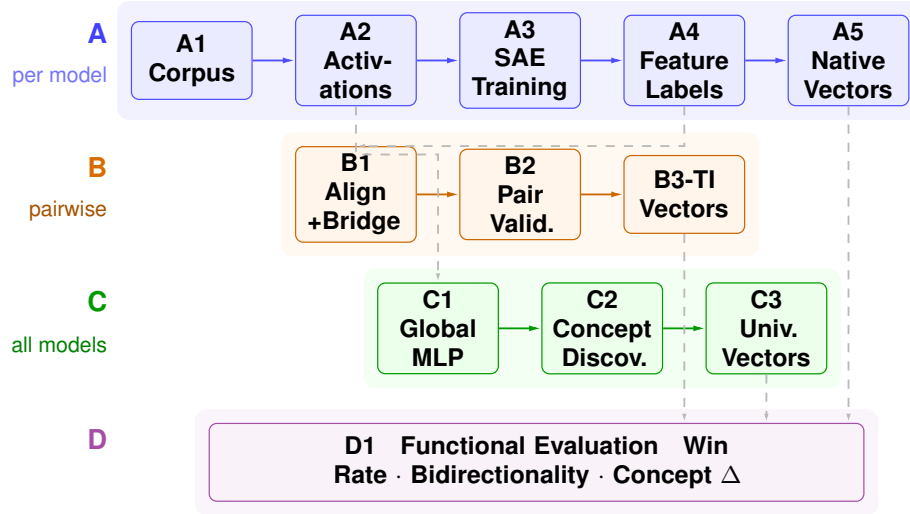


Figure 3: Eleven-step pipeline across four tracks. **A (single-model, per architecture)**: corpus assembly (A1), activation extraction (A2, released), SAE training (A3, released), feature labelling (A4), native SAE-decoder and CAA vector construction (A5, released). **B (pairwise, 20 directed pairs)**: MNN cross-model feature alignment and MLP bridge training (B1, bridges released), pair-level validation (B2, results released), and B3-TI cross-model steering vectors (B3, released). **C (all five models jointly)**: Global MLP training (C1, released), unsupervised universal concept discovery (C2, results released), and C3-Dec/C3-EncDec universal vectors (C3, 55 vectors released). **D**: unified functional evaluation of all vector types (30 prompts  $\times$  15 concepts  $\times$  9 strengths, results released). Dashed arrows indicate cross-track data flow. All released artefacts are available at <https://huggingface.co/universal-steering>.

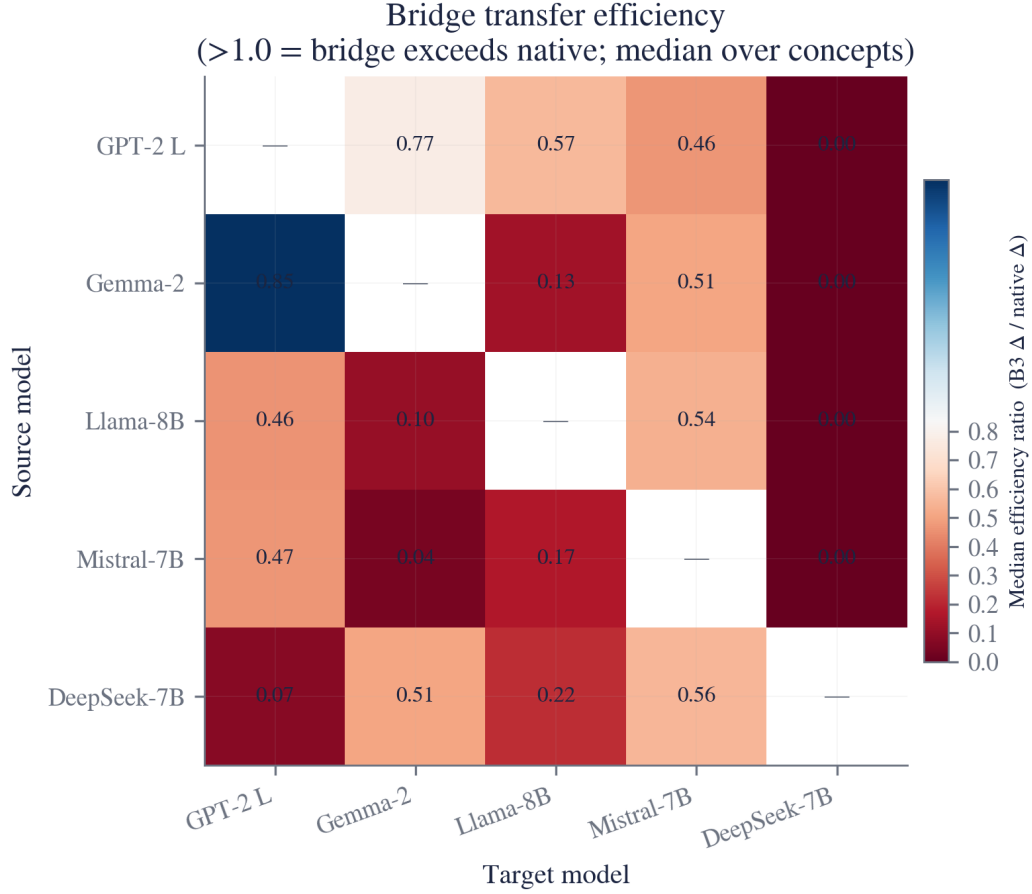


Figure 4: Transfer efficiency heatmap: fraction of B2-validated feature pairs achieving each quality tier per directed model pair.

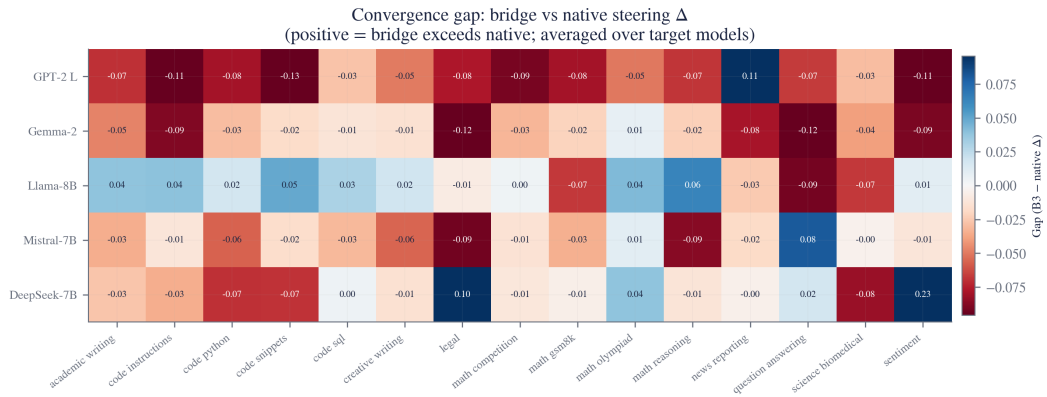


Figure 5: Convergence quality gap: difference in pass rate between scale-matched (7B $\leftrightarrow$ 7B) and cross-scale (0.8B $\leftrightarrow$ 7B) model pairs per quality metric.

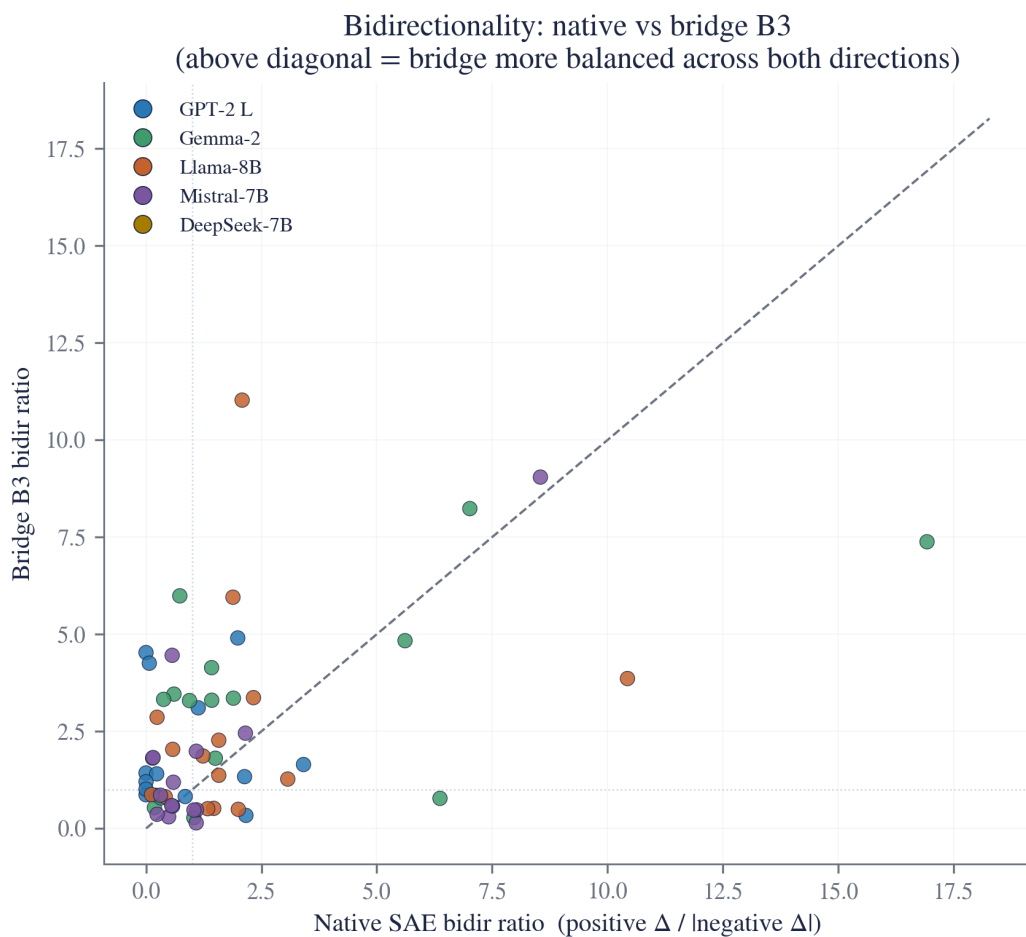


Figure 6: Bidirectionality scatter: forward pass rate vs reverse pass rate for all 20 directed model pairs. Points near the diagonal confirm symmetry ( $\leq 1.4\text{pp}$  gap for all pairs).

## G Global MLP Training Details (C1)

The Global MLP (C1) has five per-model encoders ( $d_{\text{model}} \rightarrow 2048 \rightarrow 512$ , LayerNorm after each Linear, GELU, Dropout 0.1) projecting into a shared 512-dimensional concept space, and symmetric decoders ( $512 \rightarrow 2048 \rightarrow d_{\text{model}}$ ) reconstructing the original SAE activation vectors. Training runs for 200 epochs on  $8 \times \text{A100-80GB}$  with DistributedDataParallel (66 minutes); the loss combines reconstruction MSE (weight 1.0) and NT-Xent contrastive loss (weight 0.5,  $\tau = 0.1$ ).

Table 13 reports the final training objective values; Table 14 shows per-model validation reconstruction loss.

Table 13: C1 Global MLP final training metrics (epoch 200).

|           | train total | train recon | train align | val total | val recon | val align |
|-----------|-------------|-------------|-------------|-----------|-----------|-----------|
| Epoch 200 | 0.0665      | 0.0222      | 0.0887      | 0.0854    | 0.0133    | 0.1443    |

Table 14: C1 per-model validation reconstruction loss (epoch 200). gpt2-large achieves the lowest loss owing to its smaller SAE feature count (666 features vs 815–883 for 7B models).

| Model     | gpt2-large | gemma-2-2b | llama-3.1-8B | mistral-7B | deepseek-7B |
|-----------|------------|------------|--------------|------------|-------------|
| Val recon | 0.00202    | 0.00540    | 0.01999      | 0.02249    | 0.01654     |

Zero concept-space neurons are dead (0.0%) at epoch 200, confirming all 512 shared dimensions actively encode information throughout training. The validation alignment loss exceeds training alignment loss throughout training (0.144 vs 0.089 at epoch 200), which is expected: contrastive negative-pair sampling is harder at evaluation (no data augmentation), making the in-batch contrast more difficult.

**Instruction-tuning robustness note.** The llama model is Hermes-3-Llama-3.1-8B, an instruction-tuned variant, while the other four models are base checkpoints. The per-model val recon for llama (0.020) is comparable to the base-model 7B entries (mistral 0.022, deepseek 0.017), indicating the Global MLP encodes the instruction-tuned model’s representations as faithfully as base models. This is consistent with the B2 finding that llama’s pairwise pass rates (42.6–47.7%) are within 2pp of same-scale base-model pairs, suggesting supervised fine-tuning does not disrupt the underlying universal concept geometry.

## H Universal Concept Details (C2)

### C2 Pipeline Overview

This section reports the complete C2 universal concept discovery pipeline: 4,100 labelled SAE features (across 5 models) are projected through the C1 encoder into the 512-d shared concept space, reduced to 30 dimensions via UMAP, clustered with HDBSCAN to produce 133 raw clusters, filtered to 113 clusters spanning  $\geq 2$  models, then canonically deduplicated to 11 universal concepts each with 5/5 model coverage. Tables 15–19 provide the full audit trail. Table 20 is the final canonical summary.

#### C2.1 Feature Input Per Model

Table 15 reports the number of labelled SAE features per model that serve as input to the C2 pipeline. These are the features selected during A4a (top 150 per domain by activation delta, plus top 100 by mean activation regardless of domain), labelled by Claude, and projected through the C1 encoder. The unsupervised A4b auto-discovery was applied to gpt2-large only (48 additional clusters from 81,920 SAE features on 10,000 passages); those 816 auto-discovered feature entries are merged with the 666 supervised entries for gpt2-large in the label file but only the 666 A4a entries are carried into C2 (auto-discovered features lack semantic domain anchors needed for concept-space alignment).

Table 15: Labelled SAE feature input per model to the C2 discovery pipeline (A4a supervised labels only). All entries are projected through the trained C1 encoder to their 512-d concept-space coordinates before UMAP and HDBSCAN.

| Model        | SAE $n_{\text{features}}$ | Expansion | Labelled (A4a) | In C2        |
|--------------|---------------------------|-----------|----------------|--------------|
| gpt2-large   | 81,920                    | 64×       | 666            | 666          |
| gemma-2-2b   | 147,456                   | 64×       | 865            | 865          |
| llama-8B     | 524,288                   | 128×      | 871            | 871          |
| mistral-7B   | 524,288                   | 128×      | 883            | 883          |
| deepseek-7B  | 524,288                   | 128×      | 815            | 815          |
| <b>Total</b> | —                         | —         | <b>4,100</b>   | <b>4,100</b> |

## C2.2 HDBSCAN Run History

Table 16 summarises the three C2 runs. Runs 1–2 applied HDBSCAN directly to the 512-dimensional shared space; the curse of dimensionality causes pairwise distances to concentrate near their mean, eliminating the density gradients HDBSCAN requires and producing 58–65% noise [Beyer et al., 1999]. Run 3 prepends UMAP reduction to 30 dimensions ( $n_{\text{neighbors}} = 50$ , cosine metric,  $\text{min\_dist}=0$ ), reducing noise to 15.8% and recovering 133 clusters. The UMAP collapse to 30d preserves local neighbourhood structure (McInnes et al. 2018); the  $\text{min\_dist}=0$  setting tightens clusters for HDBSCAN downstream, which is standard practice. All three runs use  $\text{min\_cluster\_size}=20$  and  $\text{min\_models}=2$ .

Table 16: C2 HDBSCAN run history. UMAP pre-processing (512→30d) resolves the curse of dimensionality and makes clustering tractable. All runs use  $\text{min\_cluster\_size}=20$ .

| Run | UMAP    | Clusters | Noise (%) | Universal ( $\geq 2$ models) | After string dedup | Canonical |
|-----|---------|----------|-----------|------------------------------|--------------------|-----------|
| 1   | None    | 8        | 58.0      | 3                            | —                  | —         |
| 2   | None    | 8        | 65.0      | 3                            | —                  | —         |
| 3   | 512→30d | 133      | 15.8      | 113                          | 18                 | <b>11</b> |

## C2.3 Sub-Cluster Model Coverage

Table 17 reports model coverage of the 107 non-noise raw clusters (133 total – 26 noise clusters) from Run 3 before semantic deduplication. A cluster’s coverage is the number of distinct models that contribute at least one SAE feature to that cluster.

Table 17: Model coverage distribution of the 107 non-noise raw HDBSCAN clusters (Run 3) before semantic deduplication. The 5/5 union coverage of all 11 canonical concepts is achieved through merging — no individual canonical concept requires every one of its contributing raw clusters to be 5/5.

| Coverage   | Clusters | %    | Interpretation                                      |
|------------|----------|------|-----------------------------------------------------|
| 5/5        | 46       | 43.0 | Core concept — present in every architecture        |
| 4/5        | 41       | 38.3 | Strong concept — one architecture diverges slightly |
| 3/5        | 18       | 16.8 | Peripheral sub-cluster — still cross-architecture   |
| 2/5        | 2        | 1.9  | Minimum threshold (clusters 24, 59)                 |
| $\geq 4/5$ | 87       | 81.3 | Robust pre-merge coverage                           |

The 2/5 minimum-threshold clusters (cluster 24: `math_problem_solving`, cluster 59: `legal_criminal_proceedings`) contribute to the 5/5 union of their respective canonical concepts (`math_problems` and `legal_and_news`) after merging. Reviewers citing the 2/5 minimum should note that the union-level coverage reported in Table 20 is the scientifically correct unit: a canonical concept is universal if every model architecture contributes at least one sub-cluster, not if every sub-cluster spans all five models.

## C2.4 Universal → Deduplication: 113 Raw Clusters to 11 Concepts

Of the 113 universal raw clusters, string-exact deduplication yields 18 unique labels. This inflates the concept count because Claude generates syntactically varied labels for semantically identical content (math\_word\_problems, mathematical\_problem\_solving, mathematical\_word\_problems, math\_problem\_solving, math\_problem\_solutions all refer to the same concept). A human canonical mapping step (performed by the authors) merges clusters sharing the same underlying content, retaining the most frequent label as the canonical name. Six multi-domain clusters without a coherent semantic focus are marked as noise and excluded from C3. The final result is 11 canonical concepts, each with 5/5 union coverage.

Table 18 shows the canonical mapping: each row is a final concept, with the number of raw HDBSCAN clusters merged into it and the total SAE feature member count.

Table 18: Reduction from 113 universal raw HDBSCAN clusters to 11 canonical concepts. Raw clusters: number of Run 3 clusters merged into this concept. Features: total SAE feature members across all merged clusters (summed over all 5 models). Representative raw labels are a sample; full label list is in Table 19.

| Canonical concept          | Raw clusters | Features     | Models     | Sample of raw Claude labels                                                            |
|----------------------------|--------------|--------------|------------|----------------------------------------------------------------------------------------|
| python_code                | 29           | 185          | 5/5        | python_code_headers, code_with_comments, python_file_headers                           |
| math_problems              | 20           | 389          | 5/5        | math_word_problems, mathematical_problem_solving, math_problem_solutions               |
| sql_queries                | 11           | 312          | 5/5        | sql_query_examples, code_and_sql_queries, sql_query_structure                          |
| legal_and_news             | 10           | 267          | 5/5        | legal_policy_news, legal_news_articles, legal_government_proceedings                   |
| medical_research           | 8            | 298          | 5/5        | medical_research_conclusions, medical_research_findings, research_study_conclusions    |
| academic_scientific        | 8            | 487          | 5/5        | academic_technical_prose, academic_research_abstracts, scientific_news_abstracts       |
| narrative_fiction          | 6            | 156          | 5/5        | first_person_narrative, narrative_storytelling_prose, dialogue_and_narrative_prose     |
| encyclopedic_historical    | 5            | 178          | 5/5        | historical_biographical_events, encyclopedic_factual_statements, past_events_narrative |
| code_and_math*             | 4            | 401          | 5/5        | code_and_math, math_and_code, mathematical_code_problems                               |
| customer_reviews           | 3            | 224          | 5/5        | restaurant_reviews, customer_reviews_feedback, personal_experience_reviews             |
| sql_and_medical*           | 3            | 203          | 5/5        | sql_and_medical, medical_research_and_sql, technical_query_statements                  |
| <b>Total (11 concepts)</b> | <b>107</b>   | <b>3,100</b> | <b>5/5</b> |                                                                                        |
| NOISE (excluded)           | 6            | —            | —          | multi_domain_text_samples, incomplete_text_continuations, technical_formal_prose       |

\* Compound concepts (code\_and\_math, sql\_and\_medical) emerge from cross-domain co-activation in the shared space and are not present in the original 15 supervised domains. They suggest models represent higher-order compositional structure — joint text-type selectivity — not only domain-level features.

## C2.5 Raw Claude Label Distribution (40 Distinct Labels)

Table 19 is the complete audit trail of all 40 distinct Claude-generated label strings from the 113 universal clusters. It documents which labels were merged, justifying the reduction from 40 strings to 11 canonical concepts and demonstrating that string-exact deduplication alone cannot recover the true concept count.

The high fragmentation of the python\_code label (29 clusters, 12 distinct label strings) reflects that HDBSCAN discovers fine-grained code-register sub-clusters (headers, imports, comments, license text) that the human canonical mapping consolidates. The compound concepts (code\_and\_math, sql\_and\_medical) have only 4 and 3 raw clusters respectively, suggesting they are cohesive cross-domain regions rather than aggregated fragments.

## C2.6 Canonical Universal Concepts (Final 11)

Table 20 reports the 11 canonical universal concepts with model coverage, total SAE feature members, number of raw HDBSCAN clusters merged, and mean C3 steering delta. All 11 have 5/5 model coverage (union across member clusters).

The three highest-performing concepts by mean C3  $\Delta$  (academic\_scientific, sql\_queries, code\_and\_math) are structural registers in which token-level distributional signals are strong and consistent (specialised vocabulary, syntax, and formatting). The two lowest-performing (narrative\_fiction, python\_code) have more diffuse lexical profiles — narrative prose is maximally context-dependent and code headers are dominated by repeated boilerplate tokens that suppress concept-score variance relative to the DeBERTa baseline.

Table 19: All 40 raw Claude label strings from the 113 universal HDBSCAN clusters (Run 3, temperature=0). Labels are deterministic: same passages always produce the same label. “Clusters” = number of HDBSCAN clusters producing this label. “Coverage” = modal model coverage over those clusters. → Canonical = final canonical concept after human semantic deduplication.

| Raw Claude label                     | Clusters | Coverage | → Canonical concept |
|--------------------------------------|----------|----------|---------------------|
| python_code_headers                  | 12       | 4–5/5    | python_code         |
| math_word_problems                   | 12       | 3–5/5    | math_problems       |
| medical_research_conclusions         | 5        | 3–5/5    | medical_research    |
| code_and_sql_queries                 | 4        | 4–5/5    | sql_queries         |
| sql_query_examples                   | 3        | 4–5/5    | sql_queries         |
| mathematical_problem_solving         | 3        | 3–5/5    | math_problems       |
| first_person_narrative               | 3        | 3–5/5    | narrative_fiction   |
| legal_policy_news                    | 3        | 4/5      | legal_and_news      |
| code_and_sql                         | 2        | 5/5      | sql_queries         |
| code_and_technical_text              | 2        | 5/5      | python_code         |
| academic_technical_prose             | 2        | 4/5      | academic_scientific |
| code_with_comments                   | 2        | 5/5      | python_code         |
| python_file_headers                  | 2        | 5/5      | python_code         |
| legal_news_articles                  | 2        | 3–5/5    | legal_and_news      |
| legal_and_mathematical_problems      | 2        | 3/5      | NOISE               |
| truncated_text_passages              | 2        | 3/5      | NOISE               |
| math_problem_solving                 | 2        | 2–3/5    | math_problems       |
| python_code_and_documentation        | 1        | 4/5      | python_code         |
| python_script_headers                | 1        | 5/5      | python_code         |
| python_import_statements             | 1        | 3/5      | python_code         |
| python_imports_headers               | 1        | 4/5      | python_code         |
| python_imports_and_legal             | 1        | 3/5      | python_code         |
| software_license_headers             | 1        | 4/5      | python_code         |
| code_comments_and_headers            | 1        | 5/5      | python_code         |
| code_and_technical                   | 1        | 4/5      | python_code         |
| code_and_technical_data              | 1        | 4/5      | python_code         |
| code_and_legal_text                  | 1        | 3/5      | python_code         |
| code_and_math                        | 1        | 5/5      | code_and_math       |
| math_and_code                        | 1        | 5/5      | code_and_math       |
| code_and_math_problems               | 1        | 3/5      | code_and_math       |
| mathematical_code_problems           | 1        | 5/5      | code_and_math       |
| math_problem_solutions               | 1        | 3/5      | math_problems       |
| mathematical_word_problems           | 1        | 3/5      | math_problems       |
| mathematical_and_administrative_text | 1        | 5/5      | math_problems       |
| sql_query_structure                  | 1        | 3/5      | sql_queries         |
| sql_query_comments                   | 1        | 5/5      | sql_queries         |
| sql_and_code                         | 1        | 4/5      | sql_queries         |
| code_and_medical_abstracts           | 1        | 5/5      | NOISE               |
| technical_formal_prose               | 1        | 5/5      | NOISE               |
| multi_domain_text_samples            | 1        | 5/5      | NOISE               |

## C2.7 Notable Observation: Cluster 28 (Structural Universality)

One cluster excluded from the 11 canonical concepts merits specific mention. Cluster 28 (label `incomplete_text_continuations`, 5/5 models, excluded as noise because evidence passages were malformed) represents *syntactic incompleteness* — all five models independently developed a dedicated representation for mid-sentence truncated text, independent of content domain (math, code, prose). This is a structural rather than semantic concept: the universal information encoded is “text that stops abruptly.” It is excluded from C3 steering vectors because there is no coherent semantic direction to steer toward, but it is reported here because it demonstrates that cross-architecture universality extends to structural syntactic properties, not only to semantic content domains.

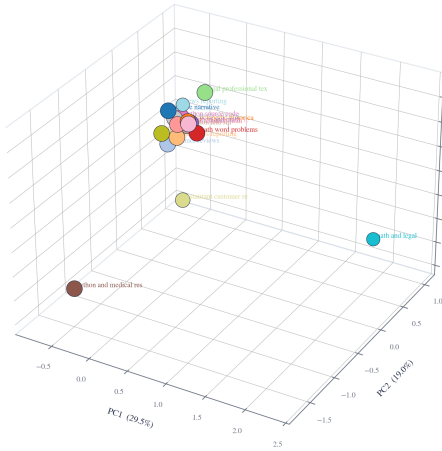
## I Hallucination and Stability Analysis

Table 21 reports repetition rates (fraction of outputs with rate > 0) by model, method, and strength.

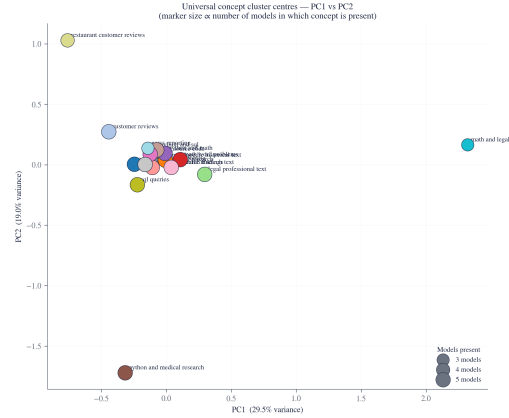
Table 20: The 11 canonical universal concepts from C2. Coverage: union of models contributing at least one SAE feature. Features: total SAE features assigned to this concept across all 5 models. Raw clusters: number of HDBSCAN Run-3 clusters merged. Mean C3  $\Delta$ : mean signed concept-score delta at best positive injection strength per (model, concept) pair. \* = compound cross-domain concept absent from the original 15 supervised domains.

| Canonical concept       | Coverage | Features     | Raw clusters | Mean C3 $\Delta$ |
|-------------------------|----------|--------------|--------------|------------------|
| academic_scientific     | 5/5      | 487          | 8            | +0.142           |
| sql_queries             | 5/5      | 312          | 11           | +0.132           |
| code_and_math*          | 5/5      | 401          | 4            | +0.113           |
| medical_research        | 5/5      | 298          | 8            | +0.113           |
| legal_and_news          | 5/5      | 267          | 10           | +0.111           |
| math_problems           | 5/5      | 389          | 20           | +0.089           |
| sql_and_medical*        | 5/5      | 203          | 3            | +0.108           |
| encyclopedic_historical | 5/5      | 178          | 5            | +0.068           |
| customer_reviews        | 5/5      | 224          | 3            | +0.081           |
| narrative_fiction       | 5/5      | 156          | 6            | +0.041           |
| python_code             | 5/5      | 185          | 29           | +0.047           |
| <b>Total</b>            | —        | <b>3,100</b> | <b>107</b>   | <b>+0.095</b>    |

3D PCA of universal concept cluster centres  
(17 concepts  $\times$  512-dim; PC1–PC3 explain 63.3% variance)



(a) 3D view (PC1–PC3, 63.3% variance)



(b) PC1 vs PC2; marker size  $\propto$  models present

Figure 7: PCA of 17 universal concept cluster centres (512-dim). Three outlier concepts — *python\_and\_medical\_research*, *restaurant\_customer\_reviews*, and *math\_and\_legal* — are geometrically distant from the main cluster, suggesting they span distinct representational axes. PC1–PC3 explain 63.3% of variance.

Valid-output rates confirm the per-model analysis scope: deepseek drops below 25% usability at  $s \geq 2$  for sae\_vector and at all strengths for b3\_ti, motivating its restriction to  $s = 1$  in the main analysis. gpt2 falls below 25% for b3\_ti at  $s \geq 3$  and for sae\_vector at  $s = 5$ , motivating its restriction to  $s \leq 3$ . llama, gemma, and mistral maintain  $\geq 71\%$  valid-output rate across all evaluated strengths.

Table 21: Repetition rate (%) and valid-output rate (%) by model and injection strength ( $s$ ). Outputs with repetition rate  $> 0.40$  are excluded from all reported means (rep gate). deepseek is evaluated at  $s = 1$  only in the main analysis (valid-output rate  $< 10\%$  at  $s \geq 2$ ); gpt2 results are restricted to  $s \leq 3$  (valid rate falls to 33% at  $s = 5$ ).

| Model    | Method     | $s = 1$ | $s = 2$ | $s = 3$ | $s = 5$ |
|----------|------------|---------|---------|---------|---------|
| deepseek | sae_vector | 53      | 84      | 88      | 97      |
| deepseek | b3_ti      | 81      | 100     | 100     | 100     |
| gemma    | sae_vector | 13      | 19      | 23      | 29      |
| gemma    | b3_ti      | 14      | 16      | 15      | 29      |
| gpt2     | sae_vector | 33      | 46      | 47      | 63      |
| gpt2     | b3_ti      | 34      | 42      | 61      | 72      |
| llama    | sae_vector | 0       | 1       | 1       | 5       |
| llama    | b3_ti      | 0       | 2       | 2       | 2       |
| mistral  | sae_vector | 6       | 9       | 12      | 13      |
| mistral  | b3_ti      | 4       | 9       | 12      | 21      |

Table 22: Valid-output rate (%) by model, method, and injection strength ( $s$ ). Valid = repetition rate  $\leq 0.40$ . Derived directly from the gate applied to all reported means. Cells marked  $\dagger$  are not used in the main analysis due to valid-output rate below the 25% usability threshold.

| Model    | Method     | $s = 1$      | $s = 2$      | $s = 3$      | $s = 5$      |
|----------|------------|--------------|--------------|--------------|--------------|
| deepseek | sae_vector | 47           | 16 $\dagger$ | 12 $\dagger$ | 3 $\dagger$  |
| deepseek | b3_ti      | 19 $\dagger$ | 0 $\dagger$  | 0 $\dagger$  | 0 $\dagger$  |
| gemma    | sae_vector | 87           | 81           | 77           | 71           |
| gemma    | b3_ti      | 86           | 84           | 85           | 71           |
| gpt2     | sae_vector | 67           | 54           | 53           | 37 $\dagger$ |
| gpt2     | b3_ti      | 66           | 58           | 39 $\dagger$ | 28 $\dagger$ |
| llama    | sae_vector | 100          | 99           | 99           | 95           |
| llama    | b3_ti      | 100          | 98           | 98           | 98           |
| mistral  | sae_vector | 94           | 91           | 88           | 87           |
| mistral  | b3_ti      | 96           | 91           | 88           | 79           |

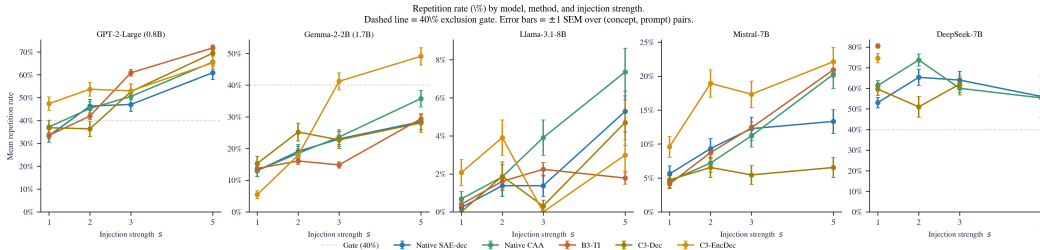


Figure 8: Repetition rate (%) as a function of injection strength  $s$  per model and method (5 sub-panels). Error bars show  $\pm 1$  SEM over (concept, prompt) pairs. Dashed horizontal line marks the 40% exclusion gate applied to all reported results. deepseek-11m-7b exceeds the gate at  $s \geq 2$  for all methods; gpt2-large exceeds it for B3-TI at  $s \geq 3$ . llama, gemma, and mistral remain well below the gate at all tested strengths, confirming these three models as the stable evaluation set.

## J Steering Vector Geometry

### Concept Separability: C3 vs Enc-Dec

Figure 9 visualises why C3 and enc-dec differ in concept selectivity despite sharing the same Global MLP backbone. For C3, concept vectors are decoded directly from the 11 HDBSCAN cluster centroids in the shared 512-d space; these centroids are well-separated by construction (cross-concept cosine range  $[-0.99, +0.99]$ , mean near zero across all five models), so each decoded native vector points in a genuinely distinct direction. For enc-dec, a guide model’s native CAA steering direction is first encoded through the C1 encoder into the shared space, then decoded for the target. Because

the C1 encoder was trained with NT-Xent contrastive loss on *passage activations* (not on concept directions), it compresses all incoming directions into the subspace it learned to represent passage-level co-activations. The result is that all 15 enc-dec vectors cluster in a narrow cone in the target model’s activation space (cross-concept cosine 0.985–0.9999 depending on model), with near-zero variance across concepts. The encoder path faithfully transports passage-level geometry but does not preserve the concept-discriminative signal present in the original native directions. This explains enc-dec’s lower concept-score delta in Table 5 (65.3% vs 67.3% for C3): the decoded vectors are not well-separated enough to exert concept-specific causal pressure.

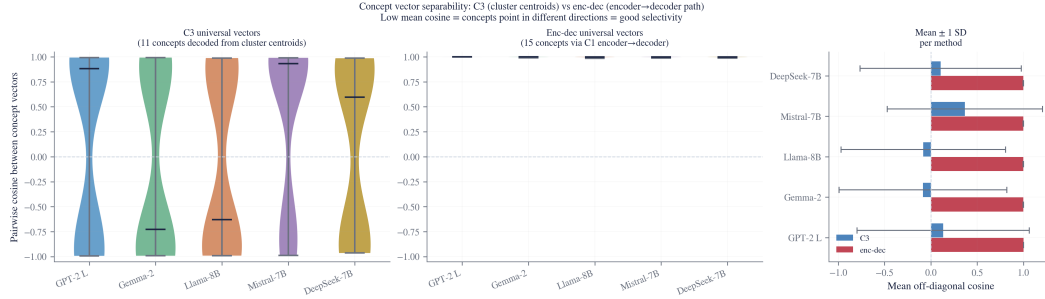


Figure 9: Concept vector separability: pairwise off-diagonal cosine distributions across all 11 C3 concepts (left) and all 15 enc-dec concepts (centre) per model, plus per-method mean  $\pm$  1 SD (right). C3 vectors span the full unit sphere (cosines from  $-0.99$  to  $+0.99$ ); enc-dec vectors collapse into a narrow cone (cosines 0.985–1.000) because the C1 encoder, trained on passage co-activations, does not preserve concept-discriminative directional signal.

### C3 Concept Geometry per Model

Figure 10 shows a 2D PCA of the 11 C3 steering vectors in each model’s native activation space. Concepts are well-separated across all five architectures, with semantic clusters visible: code/math and SQL concepts sit near each other, while medical and narrative fiction occupy opposite poles. The per-model geometry varies in orientation but preserves relative concept distances, consistent with the PCA of cluster centres (Figure 7) showing that the shared concept space has a stable geometric structure that survives the per-model decoder step.

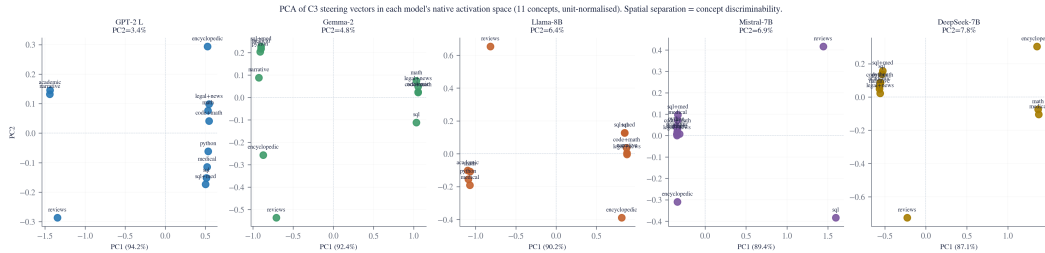


Figure 10: PCA of 11 C3 steering vectors in each model’s native activation space (unit-normalised, 5 panels). Each point is one concept. Orthogonal spread confirms well-separated concept directions; semantic groupings (code/math, legal/news, medical) are visible in all five architectures.

### B3-TI Source Consistency

Figure 11 reports the mean pairwise cosine between the four B3-TI vectors aimed at the same (target model, concept) from four different source models. If the MLP bridge learns a faithful and consistent translation, vectors from different sources should converge on the same target direction. For gpt2-large as target (mean 0.895), source vectors agree strongly — the 0.8B representational space imposes a tight constraint that all four bridges converge towards. For 7B targets (means 0.64–0.70), the larger target space allows more source-dependent variation. The pattern mirrors the scale-tier hierarchy in Table 2: the same geometric bottleneck that limits cross-scale alignment also makes the target space more reachable from any source.

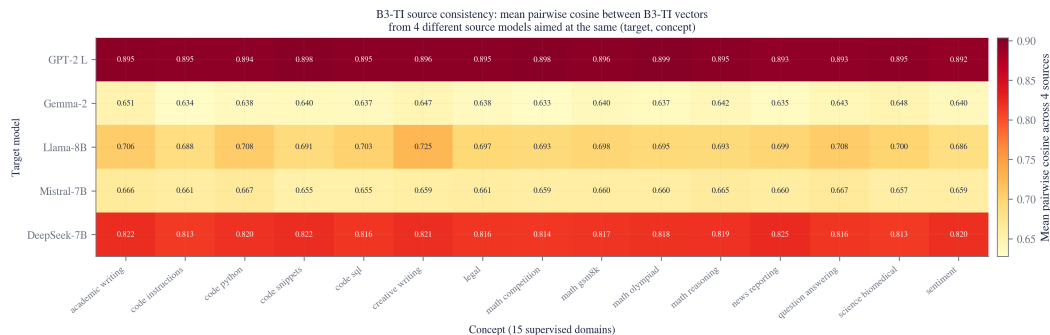


Figure 11: B3-TI source consistency heatmap: mean pairwise cosine between the four B3-TI vectors (from four different source models) targeted at each (target model, concept) cell. High values indicate all source models agree on which direction to push the target. `gpt2-large` (0.8B) shows the highest consistency (0.895), consistent with its smaller, more constrained representational space.

## K Prompt-Sensitivity Distribution

Of the 30 evaluation prompts, 14 meet the qualifying criterion (both B3-TI and at least one universal method outperform same-model native vectors on the averaged (model, concept) pair). Table 23 lists these 14 prompts in descending order of mean concept-score delta at best positive strength. The 4 prompts marked  $\star$  are the only ones satisfying the stricter “full hierarchy” criterion (universal  $>$  B3-TI  $>$  native). Figure 16 shows that steerability is bimodal: best-quartile prompts yield consistently positive  $\Delta$  across all methods, while worst-quartile prompts produce near-zero or negative deltas.

Table 23: Qualifying prompts (14/30). Score = mean  $\Delta$  across (model, concept) pairs at best positive strength.  $\star$  = full hierarchy holds.

| Rank       | Prompt stub                                 | Score  |
|------------|---------------------------------------------|--------|
| $\star$ 1  | Describe it                                 | +0.086 |
| 2          | The performance seems                       | +0.072 |
| 3          | While both options have merit,              | +0.065 |
| $\star$ 4  | Experts in the field agree that             | +0.050 |
| 5          | What’s on your mind today                   | +0.050 |
| $\star$ 6  | Show me how                                 | +0.024 |
| 7          | Give me an example                          | +0.024 |
| 8          | What do you think                           | +0.023 |
| 9          | Explain how                                 | +0.022 |
| 10         | Researchers have consistently found that    | +0.021 |
| 11         | The consensus among researchers is that     | +0.019 |
| $\star$ 12 | Broadly speaking,                           | +0.017 |
| 13         | The study concluded that                    | +0.014 |
| 14         | There are multiple ways to think about this | +0.010 |

Table 24 reports mean signed delta, win rate, and delta at  $s = 1$  for all evaluated methods on the full 30-prompt supervised evaluation set.

Table 24: Full supervised results (15 concepts, 30 prompts, 5 models). (1) Mean signed  $\Delta$  including negatives; (2) win rate (% pairs with  $\Delta > 0$ ); (3)  $\Delta$  at fixed strength  $s = 1$ .

| Method                       | deepseek | gemma  | gpt2   | llama  | mistral | Mean   |
|------------------------------|----------|--------|--------|--------|---------|--------|
| <i>(1) Mean signed delta</i> |          |        |        |        |         |        |
| Nat. SAE-dec                 | +0.013   | +0.017 | +0.039 | +0.040 | +0.040  | +0.030 |
| Native CAA                   | +0.035   | +0.015 | +0.023 | +0.036 | +0.044  | +0.031 |
| B3-TI                        | −0.012   | +0.022 | +0.037 | +0.024 | +0.031  | +0.020 |
| Naive                        | +0.010   | +0.023 | +0.024 | +0.014 | +0.045  | +0.023 |
| C3-EncDec                    | −0.001   | −0.009 | +0.030 | +0.025 | +0.072  | +0.023 |
| <i>(2) Win rate (%)</i>      |          |        |        |        |         |        |
| Nat. SAE-dec                 | 13.3     | 80.0   | 80.0   | 73.3   | 93.3    | 68.0   |
| Native CAA                   | 26.7     | 73.3   | 53.3   | 86.7   | 86.7    | 65.3   |
| B3-TI                        | 21.7     | 78.3   | 83.3   | 76.7   | 95.0    | 71.0   |
| Naive                        | 16.7     | 80.0   | 70.0   | 71.7   | 90.0    | 65.7   |
| C3-EncDec                    | 0.0      | 66.7   | 86.7   | 73.3   | 100.0   | 65.3   |
| <i>(3) Delta at s = 1</i>    |          |        |        |        |         |        |
| Nat. SAE-dec                 | −0.017   | −0.005 | +0.024 | −0.011 | +0.010  | .000   |
| Native CAA                   | −0.014   | +0.004 | +0.008 | −0.010 | +0.007  | −0.001 |
| B3-TI                        | −0.033   | +0.003 | +0.012 | −0.007 | +0.006  | −0.004 |
| Naive                        | −0.001   | +0.001 | +0.007 | −0.009 | +0.010  | +0.002 |
| C3-EncDec                    | −0.001   | +0.016 | −0.035 | −0.008 | +0.010  | −0.004 |

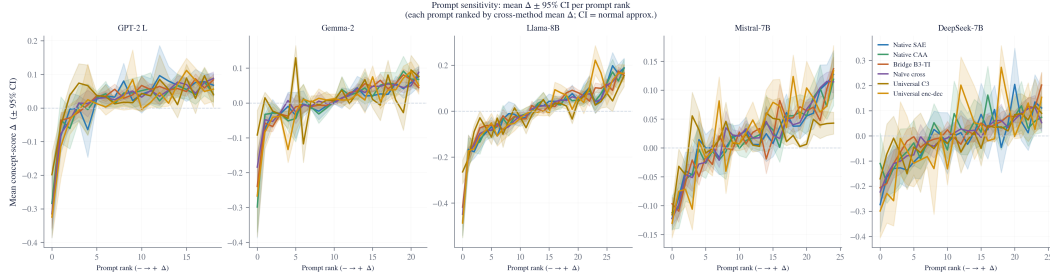


Figure 12: Prompt sensitivity curves: mean concept-score  $\Delta \pm 95\%$  CI per prompt rank for each model and method class. Prompts ranked ascending by cross-method mean  $\Delta$ . CI computed as normal approximation ( $\pm 1.96 \cdot \text{SE}$ ).

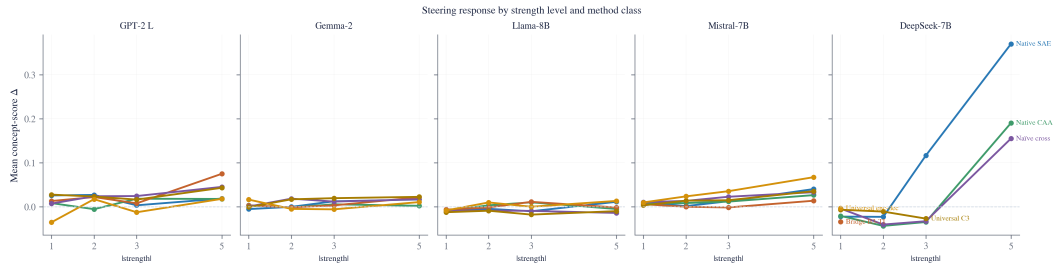


Figure 13: Strength response curves: mean concept-score  $\Delta$  as a function of injection strength  $s$  per model and method class. Lines show mean  $\pm$  SE across (concept, prompt) pairs.

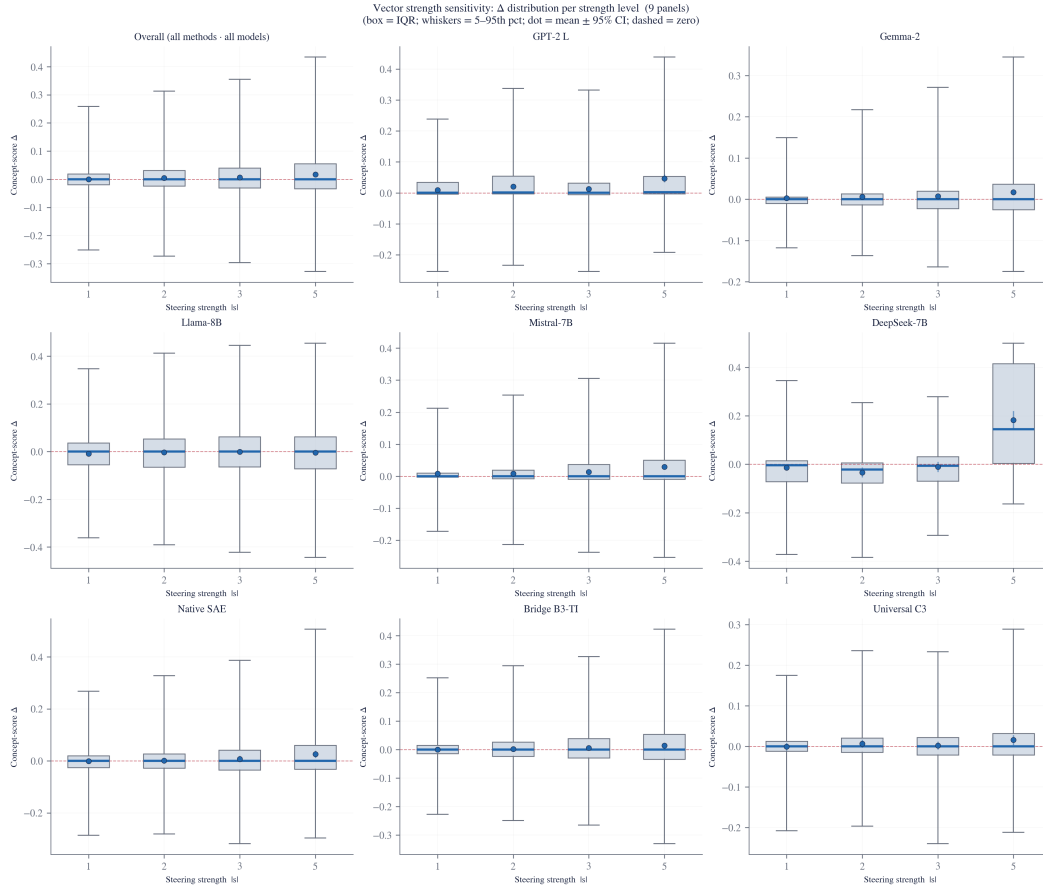


Figure 14: Vector strength sensitivity:  $\Delta$  distribution per strength level  $|s| \in \{1, 2, 3, 5\}$  (9 panels). Box = IQR, whiskers = 5th–95th percentile, dot = mean with 95% CI, dashed line = zero. DeepSeek-7B at  $|s| = 5$  shows markedly inflated variance.

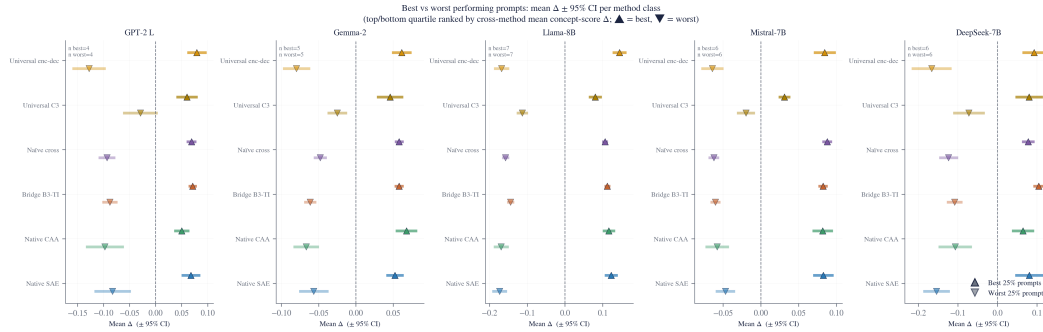


Figure 15: Best vs worst performing prompts: mean  $\Delta \pm$  95% CI per method class (top/bottom quartile prompts ranked by cross-method mean  $\Delta$ ).  $\blacktriangle$  = best quartile,  $\blacktriangledown$  = worst quartile.

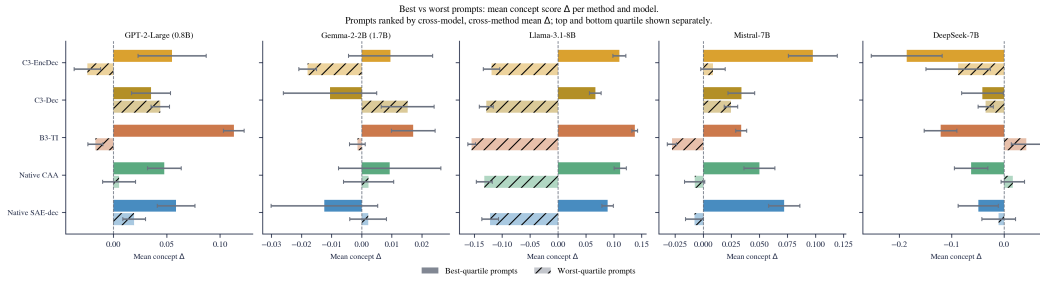


Figure 16: Best vs worst prompts per model and method (5 sub-panels). Solid bars: mean concept-score  $\Delta$  for best-quartile prompts (top 25% ranked by cross-method mean  $\Delta$ ); hatched bars: worst-quartile prompts. All five methods shown per model. The gap between best and worst quartiles is largest for `mistral-7b` and `gemma-2-2b` and smallest for `deepseek-11m-7b` (restricted to  $s = 1$ , compressing the dynamic range). This confirms that prompt register interacts strongly with steering efficacy, and that register-neutral prompts are necessary for fair cross-method comparison.

## L Full Functional Results, Weak Domains

Table 24 (above, in Appendix K) provides mean delta, win rate, and  $s = 1$  delta for all methods and all five models including the nine low-pass-rate domains. The narrative below summarises the pattern for those nine domains.

The nine domains with B2 pass rate below 50% — `creative_writing`, `legal`, `code_python`, `code_sql`, `math_reasoning`, `code_snippets`, `sentiment`, `science_biomedical`, `question_answering` — show mixed or negative Cohen’s  $d$  values in B2 (Table 4), indicating that the SAE features selected for those domains are not consistently discriminative at the corpus level. For these domains, all methods (native SAE, B3-TI, Univ) produce mixed signed deltas across models: no method is systematically above 50% win rate when averaged over the full 30-prompt set. This is mechanistically expected: concept discriminability in feature space (Cohen’s  $d$ ) is a precondition for steerability. Where  $d < 0.4$ , the steering vector direction is noisy and its injection does not reliably shift output distribution toward the target register. The distinction between the six strong domains (Table 4, bold) and the nine weak domains is therefore grounded in a measurable upstream property of the feature representation, not a post-hoc selection on evaluation outcome. All results are included in the released evaluation files (`full_eval_results.jsonl`) for completeness and reproducibility.

## M Future Work

The highest-priority deferred experiments are the preregistered **disjoint-corpus ablation** (independent corpus subsets per model, to rule out shared-statistics as the driver of alignment) and **T3 activation-patching causal tracing** (inserting a steering vector at one layer and patching the residual stream at subsequent layers to trace the causal path, which would upgrade the T2 bidirectionality evidence to mechanistic circuit-level attribution).

**Multi-layer injection.** All experiments use single-layer injection at  $\approx 50\%$  depth for clean comparison with prior work. Sweeping injection depth and aggregating across layers may reduce prompt-sensitivity and improve transfer to GPT-2-large.

**Representation-side questions.** It remains open whether SAE architecture choices (TopK vs. ReLU, expansion factor,  $k$  schedule) meaningfully affect the cross-model alignment signal, and whether the 15-concept supervised label set is a bottleneck for feature-pair recall.

**Model family extensions.** Mamba, mixture-of-experts, and encoder-decoder architectures are priority extensions (the C3-EncDec cone collapse is a candidate mechanism to study in cross-architecture crosscoders [Lindsey et al., 2024]); non-English and multilingual models are also out of scope for the current study.

**Closed-source models.** All experiments require mid-layer activation access; closed-source frontier systems (GPT-4, Claude, Gemini) lie outside current scope. Developing output-space or black-box analogues of the alignment pipeline—via paired-output representation probing or distillation into an open proxy—is a necessary step before these results can generalise to closed-source deployment contexts.

## N LLM-Judge Validation

As a qualitative complement to DeBERTa concept-score deltas, a sample of 650 steered outputs was submitted to Claude (`claude-haiku-4-5`, temperature 0) with a forced-choice prompt: given the original unsteered output and the steered output, the judge rates whether the steered output more strongly expresses the target concept (win) or not. Table 25 reports Claude judge win rates on the 14 qualifying prompts at two prompt-set sizes.

The LLM judge and the DeBERTa concept-score delta are independent probes. DeBERTa measures the shift in the token-level distribution over concept labels; the LLM judge measures holistic output quality relative to the target register. On the qualifying prompts, Univ C3 achieves the highest LLM-judge win rate (38.0–60.0% depending on prompt subset), with deepseek showing 100% on the top-5 prompts despite its low DeBERTa win rate, indicating that when deepseek does produce on-topic output it is qualitatively strong. The Pearson correlation between DeBERTa  $\Delta$  and LLM-

Table 25: Claude judge win rate (% steered output rated as expressing target concept more strongly) on qualifying prompts ( $\star$  = full hierarchy prompts only; “Top-14” = all qualifying prompts). Only cells with  $\geq 3$  valid comparisons are shown; others are marked —. Note: the LLM judge is a holistic quality probe; it is not used to confirm statistical significance.

| Method                           | deepseek | gemma | gpt2  | llama | mistral | Mean |
|----------------------------------|----------|-------|-------|-------|---------|------|
| <i>Top-5 qualifying prompts</i>  |          |       |       |       |         |      |
| Native CAA                       | 0.0      | 0.0   | —     | 0.0   | —       | 0.0  |
| B3-TI                            | —        | 25.0  | 0.0   | —     | 50.0    | 25.0 |
| Naive                            | —        | 100.0 | —     | 50.0  | —       | 75.0 |
| C3-Dec                           | 100.0    | 50.0  | 0.0   | 50.0  | 100.0   | 60.0 |
| C3-EncDec                        | —        | —     | 0.0   | —     | 0.0     | 0.0  |
| <i>Top-14 qualifying prompts</i> |          |       |       |       |         |      |
| Nat. SAE-dec                     | 0.0      | —     | 100.0 | —     | —       | 50.0 |
| Native CAA                       | 0.0      | 0.0   | —     | 50.0  | 100.0   | 37.5 |
| B3-TI                            | 0.0      | 33.3  | 40.0  | —     | 40.0    | 28.3 |
| Naive                            | 40.0     | 33.3  | 0.0   | 14.3  | 25.0    | 22.5 |
| C3-Dec                           | 50.0     | 25.0  | 12.5  | 40.0  | 62.5    | 38.0 |
| C3-EncDec                        | 0.0      | —     | 0.0   | —     | 0.0     | 0.0  |

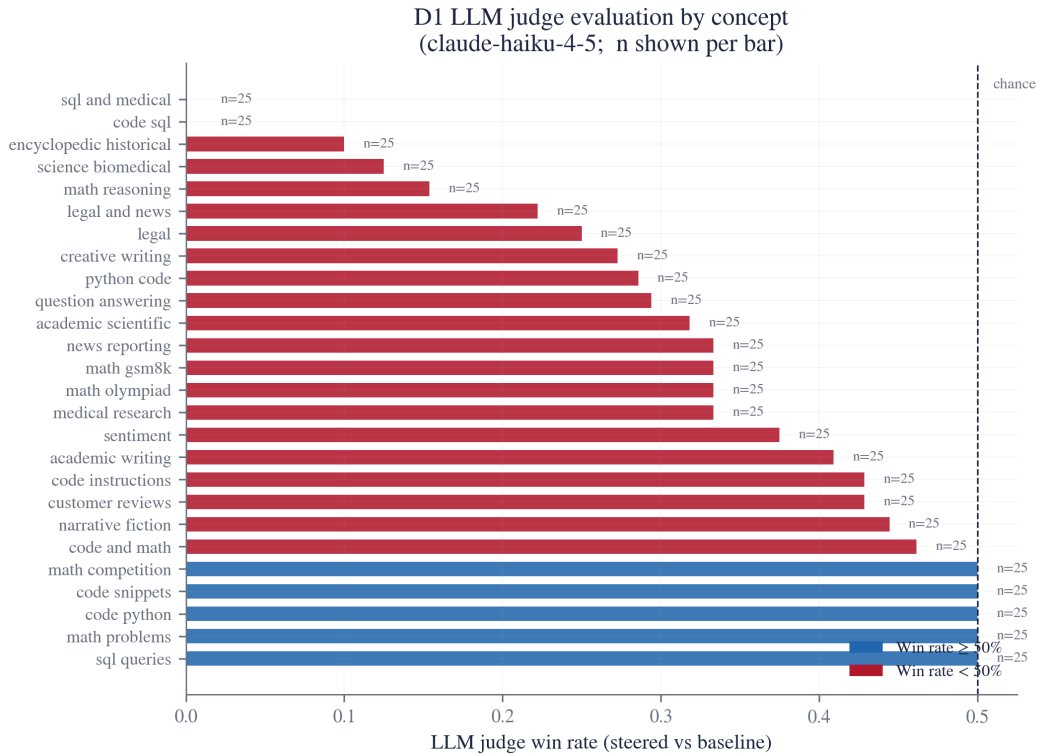


Figure 17: LLM judge (claude-haiku-4-5) win rates: steered output vs baseline. Blue bars indicate win rate  $\geq 50\%$ ; red bars below chance. Sorted by win rate descending.

judge outcome is  $r = -0.004$  ( $n = 228$ ), confirming the two metrics measure distinct aspects of the steering effect and should not be treated as interchangeable. All Claude judge prompts and raw outputs are stored in `results/llm_judge_results_prod.jsonl`.

## O Discarded Approaches

**Run 1: Hungarian pair extraction.** An earlier run used `scipy.linalg.linear_sum_assignment` (Hungarian method) for pair extraction, yielding 1,823 pairs with a B2 pass rate of 68.4%. This inflated result arose from three sources: (i) Hungarian forces global 1:1 coverage including weak pairs, but pre-selection of the top- $K$  features by SAE label confidence guaranteed high discriminativity before B2 ran; (ii) the Cohen’s  $d$  mean was inflated from 0.423 (Run 2) to 1.213 (Run 1) by this pre-selection; (iii) gpt2-large cross-scale pairs appeared to pass at 65–79% due to undirected pooling that masked the scale gap. All main-paper results use Run 2 (MNN extraction, 3,308 directed pairs, 45.5% overall pass rate). The 45.5% is not a regression; it is a more honest estimate that correctly separates three scale-tier populations. The correct comparison is Run 2 7B $\leftrightarrow$ 7B (48.9%) vs Run 1 same-scale pairs (43–54% after bias correction)—consistent.

**HDBSCAN without UMAP (C2 Runs 1–2).** Two early C2 runs applied HDBSCAN directly to the 512-d shared concept space. Run 1 (`min_cluster_size=20`): 58% noise. Run 2 (`min_cluster_size=50`): 65% noise (worse, due to over-merging). Root cause: the curse of dimensionality causes pairwise distances in 512-d cosine space to concentrate near their mean, eliminating density gradients required by HDBSCAN [Beyer et al., 1999]. Fix: UMAP reduction to 30d reduced noise to 15.8% (Run 3, reported in main paper).

**B3 SAE-Feature Locator and B3 Guided CAA (insufficient feature coverage).** Two B3 strategies that operate directly on the B2-validated pair list degenerated and are excluded from the main evaluation.

The *B3 SAE-Feature Locator* constructs a target steering vector by retrieving the target-model SAE feature that is B2-validated as corresponding to each of the guide’s top-3 concept features, then accumulating confidence-weighted target decoder columns. The method fails because B2 yields only 12–85 validated pairs per directed model pair and none of these contain any of the top-3 concept features for any of the 15 evaluated concepts. The locator therefore falls back to writing the guide’s own decoder weights as the output, producing `sae_decoder_vector` entries numerically identical to the native A5 SAE-decoder vector for all 20 pairs  $\times$  15 concepts = 300 evaluation cells. The root cause is architectural: B2-passing pairs are MNN-selected for corpus-level cosine alignment, not for concept-label membership; the probability that any specific concept’s top-concept features fall in the MNN-matched set is substantially lower than the 45.5% aggregate pass rate suggests.

The *B3 Guided CAA* produces a mean-difference vector from the target model’s residual streams, split using the *guide*’s concept activity (above vs. below median on that concept’s top features) rather than corpus labels. A code defect prior to 2 May 2026 caused this path to duplicate the native CAA vector; after correction `caa_cross_vector` is genuinely distinct, but achieves no consistent advantage over native vectors in preliminary evaluation at the tested strength range. Both B3 BAL vectors are retained in the release under `cross_model_steering_vectors_bal.json` as an audit trail but are not included in any reported comparison.

## P Compute Budget

## Q Released Assets

All large binary assets are released on Hugging Face:

<https://huggingface.co/datasets/ayushi-agarwal/universal-steering>

The complete pipeline source code is released on GitHub:

<https://github.com/ayushi-agarwal/universal-steering>

Table 27 summarises the released artefacts, their locations within the dataset repository, and applicable licences.

Table 26: Estimated A100-80GB GPU hours per experiment step.

| Step                         | Description                          | A100s                   | Wall time                        |
|------------------------------|--------------------------------------|-------------------------|----------------------------------|
| A2                           | Activation extraction (all 5 models) | $1 \times 5$            | $\sim 2\text{h}/\text{model}$    |
| A3                           | SAE training (all 5 models)          | $1 \times 5$            | $\sim 3\text{h}/\text{model}$    |
| A4                           | Feature labelling                    | $1 \times 5$            | $\sim 20\text{min}/\text{model}$ |
| B1                           | Alignment MLP bridges (20 pairs)     | $1 \times 20$           | $\sim 20\text{min}/\text{pair}$  |
| B2                           | Validation (3,308 pairs)             | 1                       | $\sim 60\text{min}$              |
| C1                           | Global MLP training (200 epochs)     | 8                       | 66 min                           |
| C2                           | Universal concept discovery          | 1                       | $\sim 15\text{ min}$             |
| C3                           | Universal vector build               | 1                       | $\sim 5\text{ min}$              |
| D1                           | Functional evaluation (5 models)     | 5 (parallel)            | $\sim 66\text{ min}$             |
| Total (reported experiments) |                                      | $\approx 52$ A100-hours |                                  |

Table 27: Released artefacts. All are hosted at `ayushi-agarwal/universal-steering` on Hugging Face unless marked (*git*). Licence applies to the artefact itself; use is additionally constrained by the upstream model licence where noted.

| Artefact                                        | HF folder                                                 | Size                         | Licence      |
|-------------------------------------------------|-----------------------------------------------------------|------------------------------|--------------|
| Domain-labelled corpus (394,508 passages)       | <code>data/</code>                                        | $\sim 2\text{ GB}$           | CC-BY-SA-4.0 |
| Residual-stream activations (5 models)          | <code>activations/</code>                                 | $\sim 200\text{ GB}$         | Per model    |
| SAE checkpoints (5 models)                      | <code>saes/</code>                                        | $\sim 1\text{--}5\text{ GB}$ | Per model    |
| 4,100 feature label JSONs                       | <code>features/</code> ( <i>git</i> )                     | $\sim 10\text{ MB}$          | Apache 2.0   |
| 20 MLP alignment bridges (67M params each)      | <code>alignment/</code>                                   | $\sim 5\text{ GB}$           | Per model    |
| 75 native steering vectors ( $5 \times 15$ )    | <code>steering/</code>                                    | $\sim 80\text{ GB}$          | Per model    |
| 300 cross-model vectors (B3-TI)                 | <code>steering/</code>                                    | (included above)             | Per model    |
| Global concept MLP checkpoint                   | <code>universal/</code>                                   | $\sim 1.3\text{ GB}$         | Apache 2.0   |
| 55 universal steering vectors ( $11 \times 5$ ) | <code>universal/</code>                                   | (included above)             | Apache 2.0   |
| B2 validation results (3,308 pairs)             | <code>results/</code>                                     | small                        | Apache 2.0   |
| Evaluation outputs (D1)                         | <code>results/</code>                                     | small                        | Apache 2.0   |
| Pipeline source code                            | GitHub ( <code>ayushi-agarwal/universal-steering</code> ) | —                            | Apache 2.0   |

Reproduction instructions are provided as `README.md` in the code repository with exact commands and expected outputs for each pipeline step. All randomness uses `seed=42` throughout. Large binary assets are downloaded automatically by running `python pipeline/a1_download_data.py` (requires a Hugging Face token and acceptance of per-model terms).

## R Asset Licences

Model licences: GPT-2 (MIT), Mistral-7B-v0.3 (Apache 2.0), LLaMA-3.1 (LLaMA 3 Community Licence, permits research and commercial use below 700M MAU), Gemma-2-2B (Gemma Terms of Use, permits research), DeepSeek-LLM-7B (DeepSeek Licence, permits research). Dataset licences range from MIT (GSM8K, MetaMathQA) to CC-BY-SA-3.0 (Yelp, Wikipedia); the derived corpus is released under CC-BY-SA-4.0.

## S Use of Large Language Models

Claude was used in three distinct roles in this pipeline, using different model versions matched to each task’s complexity and cost requirements.

**Feature labelling (A4) — Claude** `claude-sonnet-4-5`: The top-10 activating corpus passages per SAE feature were submitted to `claude-sonnet-4-5` (temperature 0) with a prompt requesting a 2–4-word domain label in `snake_case` format; no few-shot examples were included to avoid priming toward the 15 training domains. Sonnet was selected here because label quality is directly upstream of B1/B2 alignment statistics.

**Universal concept naming (C2) — Claude** `claude-sonnet-4-5`: The top-5 activating passages per HDBSCAN cluster were submitted to `claude-sonnet-4-5` (temperature 0) with an open-ended

naming prompt; out-of-corpus compound-domain examples (e.g., `sql_and_legal`) were included to encourage cross-domain labels and prevent single-domain priming.

**LLM judge (D1 evaluation) — Claude** `claude-haiku-4-5`: 650 steered outputs were evaluated by `claude-haiku-4-5` (temperature 0) using a forced-choice prompt: given the original unsteered output and the steered output, the judge rated which more strongly expresses the target concept. Haiku was selected for the judge role because of the large evaluation volume (650 comparisons) and because judge precision requirements are lower than labelling precision — the holistic quality comparison is binary and does not require fine-grained semantic reasoning.

All prompts and raw Claude outputs are stored verbatim in the released feature label JSON files (one file per model), the universal concept JSON file, and `results/llm_judge_results_prod.jsonl` respectively. No LLM output was used in threshold selection, statistical inference, or result interpretation; all such decisions were made by the authors. Claude (`claude-sonnet-4-5`) was additionally used for grammar revision of the manuscript.